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SAFE ASSET SCARCITY, COLLATERAL REUSE, AND MARKET FUNCTIONING

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Schneider

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Abstract

Quantitative easing contributes to safe asset scarcity and repo market specialness. We show that banks respond to scarcity induced by Eurosystem bond purchases by increasing their reuse of these bonds as collateral. While reuse is low, additional reuse dampens scarcity effects. However, repo rates become increasingly sensitive to asset purchases when reuse is high. Elevated reuse also impairs market functioning: it leads to more frequent and longer spells of delivery failures, a higher volatility of repo rates, and mispricing in the cash bond market. Our results highlight a tradeoff between the shock absorption and shock amplification effects of collateral reuse.

JEL Classification: E4, E5, G1, G2

Keywords: Safe assets, Government bonds, Collateral reuse, Rehypotheication, Repo market, Securities lending

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Safe asset scarcity, collateral reuse, and market functioning^{*}

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April 16, 2026

Abstract

Quantitative easing contributes to safe asset scarcity and repo market specialness. We show that banks respond to scarcity induced by Eurosystem bond purchases by increasing their reuse of these bonds as collateral. While reuse is low, additional reuse dampens scarcity effects. However, repo rates become increasingly sensitive to asset purchases when reuse is high. Elevated reuse also impairs market functioning: it leads to more frequent and longer spells of delivery failures, a higher volatility of repo rates, and mispricing in the cash bond market. Our results highlight a tradeoff between the shock absorption and shock amplification effects of collateral reuse.

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1 Introduction

Safe assets play an important role in the economy: they store value over time and serve as collateral in financial transactions. However, an increasing global demand for high-quality assets has been facing a declining pool of issuers in recent years, raising concerns about a shortage of safe assets. Large-scale purchases of sovereign debt by central banks have further contributed to a scarcity of high-quality collateral. Yet, the availability of safe assets as collateral does not only depend on the issued volume net of central bank holdings. Market participants can adjust to scarcity by reusing collateral more efficiently. In this paper, we use a unique granular dataset to study banks' reuse of specific collateral in response to safe asset scarcity induced by central bank asset purchases and its impact on bond market functioning.

When market participants receive a security as collateral in one transaction, they can reuse it to support another transaction with a different counterparty. For example, they can use the security to raise cash in a repurchase agreement ('repo') or earn a fee in a securities lending transaction. The collateral receiver, in turn, can reuse the security in another transaction, for example in a short-sale or in another repo. The number of times a piece of collateral is reused in unrelated transactions is referred to as "collateral velocity". Conceptually, the collateral velocity resembles the well-known money multiplier. As collateral can be reused multiple times, even a modest increase in collateral reuse would contribute to a significant increase in effective collateral available for market transactions. The more often a security is reused as collateral, the higher the volume of financial transactions it is backing. In theory, collateral velocity can be infinite, but in practice it is constrained by haircuts ([Bottazzi, Luque and Pascoa, 2012](#)) or other institutional constraints ([Gorton, Laarits and Metrick, 2020](#)).

The previous literature distinguishes between the scarcity of safe assets in general, and the scarcity of specific securities that are used as collateral. Regarding the former, [Krishnamurthy and Vissing-Jorgensen \(2012\)](#) among others highlight that Treasury securities

have money-like features, providing investors with “extreme safety and liquidity”. Similarly, [Gorton \(2017\)](#) refers to safe assets as a class of securities for which counterparties typically do not disagree about their fundamental value and which can be easily exchanged for other assets, with U.S. Treasuries being a prime example. [Infante and Saravay \(2025\)](#) focus on reuse as a mechanism that alleviates the scarcity of safe assets in general by redistributing their benefits across investors. The notion of scarcity of safe assets in general is in contrast to the scarcity of specific securities. For example, [Duffie \(1996\)](#) and [Krishnamurthy \(2002\)](#) discuss the implications of scarcity of a specific asset relative to other securities from the same asset class (e.g. “on-the-run” vs. “off-the-run” Treasuries). In this paper, we focus on the latter notion of safe asset scarcity, documenting that banks’ reuse of specific Bund securities may alleviate or amplify the scarcity effects induced by central bank purchases depending on how much of the asset is already being reused. In addition, we document that the effect of purchase-induced scarcity on the rate of reuse is larger for German Bunds, the sovereign debt generally perceived to be the safest in Europe.

When a specific safe asset is scarce, this may give rise to repo market specialness. A bond is referred to as trading “on special” in the repo market when the specific repo rate for that bond is lower than the rate for general collateral. The specialness spread, the difference between the repo rate for specific and general collateral, thus measures the extra cost of borrowing a specific security in the repo market. In a seminal paper, [Duffie \(1996\)](#) studies the drivers of repo market specialness. In his model, specialness spreads are increasing in the demand for short positions as well as the degree to which owners of the security are restricted from supplying it as collateral. Legal or institutional barriers can inhibit the supply and thus raise transaction costs. While he does not explicitly consider reuse, collateral reuse would increase the overall availability of collateral in the market ([Gottardi, Maurin and Monnet, 2019](#)).

In this paper, we empirically document that the prevailing degree of collateral reuse is key to understanding the impact of safe asset scarcity on specialness spreads. We make

the following contributions. First, we document that banks substantially increase their reuse in response to scarcity induced by central bank purchases. Second, we show that this reuse can mitigate scarcity premia. Importantly, though, we highlight a nonlinearity: while reuse is low, additional reuse dampens scarcity effects. On the flip side, repo rates become increasingly sensitive to asset purchases when reuse is already high. Finally, we show that additional reuse is associated with a deterioration of government bond market functioning at elevated levels of reuse: it leads to a higher rate and prevalence of cascading spells of delivery failures in the repo market, an increased volatility of repo rates, and larger mispricing in the secondary cash bond market.

The reuse of collateral is a widespread practice in the financial system. However, due to a lack of data, little is known about the extent to which collateral is reused by financial intermediaries. Very few studies quantify collateral reuse, and typically only at the dealer level, relying on dealers’ annual reports ([Singh and Aitken, 2010](#); [Singh, 2011](#)) or on supervisory data ([Infante, Press and Strauss, 2018](#); [Infante, Press and Saravay, 2020](#); [Infante and Saravay, 2025](#)). Granular data on how dealers manage their collateral portfolios at the security level has not been available thus far. We fill this gap by making use of a unique regulatory dataset that provides comprehensive *security-by-security* information on banks’ collateral positions. Specifically, we exploit a unique feature of the Bundesbank’s Securities Holdings Statistics (SHS), which provides bond-level information on the amount of collateral received and posted in securities lending and repo transactions for each security for all German dealer banks.

Central bank purchases of government bonds provide a useful laboratory to study the effects of safe asset scarcity on collateral reuse and its implications for bond market functioning. Under the Public Sector Purchase Programme (PSPP), the Eurosystem bought more than EUR 2.5 trillion of member countries’ sovereign debt between March 2015 and May 2022. These purchases corresponded to almost 30 percent of the total amount of euro area government debt outstanding and thus represented a significant

reduction of collateral available in the market. While the overall purchases for different asset classes were published by the Eurosystem, the specific purchase amounts of individual securities that we observe were unknown to market participants. This allows us to study how an exogenous reduction in available collateral affects dealers' reuse activity and what this implies for market functioning. In our main analysis, we focus on the security-level reuse of German federal government bonds (Bunds), which is the collateral most commonly used by German dealer banks. We also compare our results with a larger pool of euro area sovereign debt securities.

When a central bank purchases government debt from banks, it simultaneously issues reserves, resulting in no change to the total amount of safe assets. However, reserves and sovereign debt are not perfect substitutes for market participants demanding specific securities, for example a hedge fund pursuing a trading strategy that requires a particular bond. In the euro area, this is particularly relevant since there has been a transition from cash-driven to security-driven repo transactions in recent years ([Brand, Ferrante and Hubert, 2019](#)).¹

The rate at which dealers reuse incoming collateral is high, fluctuating between 60% and more than 95% in our sample period which covers the years 2008 through 2017. Notably, this reuse rate increased markedly after the start of Eurosystem asset purchases. The amount of collateral reused is also substantial relative to the outright ownership in these bonds, increasing from two times dealers' own holdings before the start of the PSPP to more than 25 times at the end of the sample.

¹There may be additional frictions that result in imperfect substitutability between sovereign bonds and central bank reserves. Most importantly, reserves cannot be used as collateral in securities financing transactions because they are account entries held exclusively by depository institutions at the central bank. As such, they are neither transferable nor directly accessible to non-bank financial institutions (NBFI) ([Gorton and He, 2023](#)). This is important since the NBFI sector has substantially gained market share in recent years ([Acharya, Cetorelli and Tuckman, 2024](#)). Additionally, while the safest sovereign debt such as U.S. Treasuries and German Bunds are internationally accepted safe assets held broadly by international investors ([Maggiori, Neiman and Schreger, 2020](#); [Nenova, 2025](#)), reserves are domestic to a currency area and cannot be easily ported across jurisdictions.

Using proprietary security-level information on PSPP purchases of German government bonds, we document a sizable adjustment in reuse when collateral becomes more scarce: an asset purchase of one percent of the bond’s outstanding amount increases the collateral reuse in that bond by 0.13%-0.21% in the same month. This increase is driven by two channels. First and foremost, banks increase the rate at which they reuse collateral that they already have available. Additionally, they also source more collateral from other market participants for reuse. Importantly, we control for various demand and supply effects in these regressions, including changes in the amount outstanding, securities’ on-the-run and cheapest-to-deliver status, the use of the Bundesbank’s securities lending facility, and German debt management office’s reissuance cycle.

To what degree does banks’ endogenous adjustment of collateral reuse mitigate safe asset scarcity? In line with the estimates in [Arrata, Nguyen, Rahmouni-Rousseau and Vari \(2020\)](#), we find that an asset purchase of 1% of the amount outstanding reduces Bunds’ specific collateral repo rates by about one basis point, thus making it more costly to borrow these securities. While the previous literature had documented a link between central bank purchases and scarcity premia, we show that banks mitigate this scarcity by actively increasing collateral reuse. However, their ability to do so crucially depends on the prevailing level of reuse in a given bond: higher reuse activity substantially raises the sensitivity of the repo rate to asset purchases. These results highlight the importance of collateral reuse for mitigating or amplifying asset scarcity. Repo rates are less sensitive to scarcity induced by asset purchases at low levels of reuse, and more so when reuse activity is already high and thus the possibility to raise it further is limited.

A potential side effect of high collateral reuse could be an increase in the interconnect-
edness among market participants, which in turn might contribute to an amplification of shocks in the financial system ([Brumm, Grill, Kubler and Schmedders, 2023](#); [FSB, 2017b](#)). Moreover, extended reuse may result in a concentration of collateral markets and thus contribute to fragility of dealer banks’ balance sheets ([Infante and Vardoulakis, 2021](#)).

We assess the empirical relevance of these effects by studying the link between reuse and government bond market functioning.

We find that higher reuse rates are associated with more frequent failures to deliver specific bonds, with a higher prevalence of cascading spells of delivery failures, and with increased repo rate volatility. This effect is particularly pronounced for bonds that trade on special in the repo market. The link between reuse and future repo market fails and volatility is nonlinear, however. While there is no discernible relation for reuse rates below 60%, there is a strong positive link above that threshold. Combined, these findings highlight an important tradeoff: collateral reuse can dampen scarcity effects when reuse is low, but amplify the side effects of scarcity on repo market functioning when the level of reuse is already high.

There is a close link between repo markets and government bond markets ([Duffie, 1996](#); [Jordan and Jordan, 1997](#)). As a consequence, purchase-induced scarcity may result in mispricing in the secondary market for government bonds ([D’Amico, Fan and Kitsul, 2018](#)) and a disconnect between cash and futures markets ([Pelizzon, Subrahmanyam and Tomio, 2024](#)). We document this connection also in the case of PSPP purchases. However, we show that the impact of scarcity on pricing conditions in the secondary market is again tied to the level of reuse. For elevated reuse rates asset purchases significantly amplify the effect of scarcity premia on mispricing in the cash bond market, but this is not the case for reuse rates below that threshold.

In sum, we show that banks endogenously respond to a shortage of safe assets by enhancing the reuse of collateral. At the extensive margin they seek more collateral from counterparties, and at the intensive margin they reuse available collateral at a higher rate. While the associated increase in effective collateral availability dampens the impact of asset purchases on scarcity premia, repo and cash bond market functioning deteriorate with the degree of reuse activity. Hence, we highlight an important tradeoff: reuse can

mitigate safe asset scarcity and help buffer shocks to net supply, but when it is already high collateral reuse may increase bond market fragility and act as a shock amplifier.

2 Related literature

Our paper relates to several strands of the literature. First and foremost, we contribute to the literature on safe asset shortages and its consequences for the economy. This literature distinguishes between the scarcity of safe assets in general (e.g. [Krishnamurthy and Vissing-Jorgensen, 2012](#); [Gorton, Lewellen and Metrick, 2012](#); [Gorton, 2017](#)), and the scarcity of specific securities used as collateral (e.g. [Duffie, 1996](#); [Krishnamurthy, 2002](#)). With regard to the literature on the shortage of specific securities, we document that reuse can significantly alleviate scarcity in the repo market. However, elevated levels of reuse amplify scarcity effects and are associated with a deterioration of market functioning. With regard to the broader literature on safe asset shortages, our results highlight that central bank-induced scarcity affects reuse most strongly for the safest euro area bonds.

A growing theoretical literature explores the role of collateral reuse in financial markets, generally acknowledging a tradeoff between economic efficiency and financial stability with respect to collateral reuse (e.g. [Lee, 2017](#)). [Bottazzi et al. \(2012\)](#) show that constraints on the rehypothecation of assets induce liquidity premia in repo markets and study the conditions under which a repo market equilibrium exists. [Infante \(2019\)](#) highlights that runs may arise due to collateral reuse, and [Infante and Vardoulakis \(2021\)](#) further show that such runs may generate fragility of dealer banks' balance sheets. In [Andolfatto, Martin and Zhang \(2017\)](#) reuse of collateral improves the efficient allocation of liquidity. [Gottardi et al. \(2019\)](#) show that repurchase agreements arise optimally as a borrowing instrument in competitive markets with limited commitment. Moreover, lenders reuse collateral in equilibrium, and the benefits of reuse are highest when assets are scarce. Consistently, we find that reuse rises when central bank asset purchases increase scarcity. In [Brumm et al. \(2023\)](#) moderate collateral reuse improves welfare due to more efficient

risk sharing, but excessive reuse increases leverage and volatility in the economy, reducing welfare. In line with this mechanism, we document a direct link between reuse and repo market quality.

To quantify reuse, researchers have initially resorted to dealers’ annual reports (Singh and Aitken, 2010; Singh, 2011; Kirk, McAndrews, Sastry and Weed, 2014). More recently, Infante et al. (2020) and Infante and Saravay (2025) quantify dealer-level collateral reuse activity from U.S. confidential supervisory data. Infante and Saravay (2025) analyze reuse as a mechanism for alleviating the scarcity of safe assets in general. They document that while the overall reuse of Treasury securities responds symmetrically to changes in the total supply caused by net Treasury issuance or Federal Reserve asset purchases, the sensitivity of reuse to changes in the demand for safe assets is different from the sensitivity to supply changes. Complementing their analysis, we rely on dealer-security-level data to study the compensating effect of reuse on the scarcity of specific safe assets and the implications of reuse activity on market functioning. Fuhrer, Guggenheim and Schumacher (2016) construct a measure of collateral reuse in the Swiss repo market from transaction data, showing that collateral reuse decreases with the availability of collateral.² Our dataset captures the reuse of collateral not only in the repo but also in the securities lending market, which represents an important part of the collateral intermediation chain. In the context of collateral transformation, Aggarwal, Bai and Laeven (2021) highlight the importance of the securities lending market for accessing safe assets during periods of market stress. Ferrari, Guagliano and Mazzacurati (2017) propose broker-to-broker activity in the securities lending market as a proxy for collateral reuse activities and document that it is negatively related to bonds’ specialness premia, suggesting an endogenous market reaction to scarcity. This is consistent with our finding that reuse increases in response

²Fuhrer et al. (2016) propose an algorithm to quantify collateral reuse from repo transaction data. Applying their method to the Swiss franc repo market, they find that around 5% of the interbank market was secured with reused collateral. This is a rather low level of reuse compared to the estimates from dealers’ annual reports (70-80%). The low estimate for reuse is likely due to the fact that the authors only consider repos denominated in Swiss francs and also cannot factor in securities lending transactions.

to a reduction of available high-quality collateral, and that scarcity premia are lower for securities with a higher level of reuse.

We also contribute to the literature studying the links between collateral intermediation and liquidity risks. [Luu, Napoletano, Barucca and Battiston \(2021\)](#) show that systemic liquidity risk depends on the structure of the collateral network, with longer chains being associated with more collateral velocity but also increased risk of collateral hoarding in times of stress. [Infante and Vardoulakis \(2021\)](#) document that dealers' intermediation of collateral, which often involves rehypothecation chains, may give rise to collateral runs, a form of systemic liquidity risk. Several authors ([Fleming and Garbade, 2005](#); [Garbade, Keane, Logan, Kirby and Wolgemuth, 2010](#); [Iyer and Macchiavelli, 2017](#)) have documented that cascading failures to deliver indicate funding market stress and market-wide deterioration of liquidity. We complement this literature by documenting that elevated reuse rates are associated with more frequent failures to deliver, with a higher prevalence of cascading spells of delivery failures, and with increased repo rate volatility.

Finally, we contribute to the literature on repo markets and bond specialness (e.g. [Jordan and Jordan, 1997](#); [Krishnamurthy, 2002](#)). A few studies have analyzed the scarcity effects of central bank asset purchase programs on the repo market in the U.S. and the euro area, providing evidence that such purchases raise specialness spreads ([D'Amico et al., 2018](#); [Jank and Moench, 2018](#); [Brand et al., 2019](#)). In a recent study, [Arrata et al. \(2020\)](#) also document that purchases through the ECB's Public Sector Purchase Programme (PSPP) lead to a reduction in repo rates. Moreover, [Corradin and Maddaloni \(2020\)](#) show that the ECB's Securities Markets Programme (SMP) increases the probability of fails-to-deliver. Our results document that the endogenous response of banks in using scarce collateral more effectively reduces the impact of such purchases on repo market specialness and on fails-to-deliver when reuse can easily be expanded, but that high levels of reuse amplify the sensitivity of specialness premia and delivery failures to central bank asset purchases.

3 Measuring collateral reuse

According to the broad definition provided by the [FSB \(2017b\)](#), collateral reuse includes “any use of assets delivered as collateral in a transaction by an intermediary or other collateral taker”. Market participants receive securities as collateral from various transactions, such as reverse repos, securities lending, margin lending, and over-the-counter derivative transactions. If the incoming collateral is eligible for reuse, the financial institution can reuse the security to support another such transaction. Received collateral can also be used to establish a short position. The definition of collateral reuse is more general than the narrower concept of collateral rehypothecation, which refers to the use of client’s assets as collateral.

We study banks’ incoming and outgoing collateral from securities financing transactions, which include reverse repo transactions and securities lending. Importantly, collateral received from these transactions is eligible for reuse since securities lending and repo transactions in Europe generally involve full temporary transfer of title of the underlying security. Data collected by the ESRB suggest that a large proportion of collateral reuse is currently occurring via securities financing transactions. Specifically, [Keller et al. \(2014\)](#) report that for European banks 98% and 99% of collateral received through reverse repo and securities lending/borrowing transactions are eligible for reuse. Hence, for simplicity we assume that all incoming collateral is eligible for reuse.

To compute collateral reuse at the dealer-security level we rely on the Bundesbank’s Securities Holdings Statistics (SHS) which provides security-level data on German banks’ portfolios at quarter and – since 2013 – month ends. In addition to the banks’ own holdings, the data also include for each security the amount of incoming and outgoing collateral from securities lending and repo transactions. Due to their conceptual similarity, securities lending transactions and repos are pooled in the securities holdings statistics. The original purpose of collecting figures on incoming and outgoing collateral is to avoid double counting

in securities holdings. We utilize this information to compute security-specific reuse activity at the bank level.³

We focus on German sovereign bonds with a remaining maturity between 1 and 30 years and denominated in euro. Before calculating our reuse measures we apply the following filters. We employ a plausibility check to our data by omitting positions where the outgoing collateral exceeds the sum of amount owned outright and incoming collateral. Moreover, we restrict the sample to bonds that are actively used by German banks as collateral. Specifically, we drop observations where the outgoing collateral is zero both for the current and the previous period. We also compare our baseline findings for German Bunds with a larger sample of collateral issued by euro area member countries across different ratings classes.

The upper panel of Figure 1 describes the dynamics of the aggregate incoming and outgoing collateral, normalized by the outright ownership across dealers and securities. Both ratios move in lockstep, already suggesting that much of the received collateral is reused when collateral is posted. Moreover, the figure shows that both incoming and outgoing collateral considerably exceed the dealers' outright holdings. Both metrics range between two and about 25 in our sample period. In particular, we see a strong increase in incoming and outgoing collateral after the introduction of the PSPP in March 2015. This suggests an increase in reuse activity in response to safe asset scarcity induced by central bank asset purchases.

Using banks' own holdings as well as their incoming and outgoing collateral allows us to quantify their collateral reuse activity in each security. Our main measure follows the FSB's (2017a) final recommendation for measuring reuse:

$$Reuse_{i,j} = \left(\frac{Incoming\ collateral_{i,j}}{Incoming\ collateral_{i,j} + Outright\ ownership_{i,j}} \right) \times Outgoing\ collateral_{i,j}, \quad (1)$$

³While the data do not contain any information on haircuts, they typically do not feature much variation, especially for safe assets, such as highly-rated government bonds (Gorton et al., 2020).

where *Incoming collateral*_{*i,j*} is the market value of bond *i* received as collateral by dealer *j*, *Outgoing collateral*_{*i,j*} is the market value of bond *i* posted as collateral or sold short by dealer *j*, and *Outright ownership*_{*i,j*} is the market value of dealer *j*'s outright ownership of bond *i*. The intuition is as follows. Outgoing collateral can in principle originate from two sources: own assets or incoming collateral, fulfilling the following inequality: $Outgoing\ collateral_{i,j} \leq Outright\ ownership_{i,j} + Incoming\ collateral_{i,j}$. The measure assumes proportional use of own assets and incoming collateral when posting collateral, which is captured by the first term of Equation (1). Two examples illustrate this. If incoming collateral is zero, then all outgoing collateral has to come from outright ownership and reuse for that bond is zero. In contrast, if outright ownership is zero, then all outgoing collateral has to be sourced from incoming collateral and reuse for that security is 100%. For all values in between, reuse is proportional to the incoming collateral's share of total collateral available. This proportional use of collateral is in line with the responses received by market participants to a call for evidence by the FSB. According to the FSB (2017a) survey, market participants do not generally distinguish between own securities or securities originating from another collateralized transaction when posting collateral.⁴ The upper panel of Figure 1 shows the aggregate amount of collateral reused, normalized by the outright ownership. The ratio closely tracks the corresponding ratios with incoming or outgoing collateral in the numerator. All three series show a sharp increase during the PSPP period.

⁴An important aspect of repo markets is central clearing. Central counterparties (CCPs) provide several advantages, such as netting and clearing of bilateral positions. However, CCPs may also impede collateral reuse. In Europe trading platforms and associated CCPs tend to specialise in particular segments of the market (Schaffner, Ranaldo and Tsatsaronis, 2019). Such specialization reduces the scope for netting to market participants. Our data does not provide information on banks' counterparties or the use of central clearing. But even if a fraction of dealers' collateral is excluded from reuse by CCPs, our measure captures collateral reuse that takes place despite such frictions.

To capture the intensive margin of reuse activity, we compute the rate at which dealer j reuses collateral in bond i as follows:

$$Reuse\ rate_{i,j} = \left(\frac{Reuse_{i,j}}{Incoming\ collateral_{i,j}} \right) \quad (2)$$

The reuse rate thus measures the fraction of incoming collateral that has been reused by a bank in a specific security. It indicates how extensively the bank uses its collateral resources and is therefore sometimes referred to as “collateral efficiency” (Kirk et al., 2014). When there is no incoming collateral we define $reuse\ rate_{i,j} := 0$. A substantial part of our empirical analysis relies on collateral reuse measured at the level of individual bonds. To this end, we compute the security-specific $reuse\ rate_i$ by aggregating for bond i the amount of incoming and reused collateral over all German dealer banks.

The lower panel of Figure 1 shows the aggregate reuse rate for German sovereign bonds over time. Consistent with anecdotal evidence, collateral reuse declined in times of market stress such as the global financial crisis of 2007-2008 and the European sovereign debt crisis. Moreover, there appears to be a decline in the reuse rate around 2015, coinciding with the Basel III leverage ratio disclosure requirement. After the start of the Eurosystem’s Public Sector Purchase Programme (PSPP) in 2015, reuse activity has been rising sharply. Table 1 provides summary statistics for the different reuse measures as well as further variables of interest which we use in our regression analysis.

In Appendix A, we compute two alternative measures of reuse which represent an upper and lower bound for the proportional measure of collateral reuse activity, respectively. We find that the three measures imply very similar reuse rates.⁵ In what follows, we thus focus on the proportional measure of collateral reuse since it most closely resembles actual market practices. As a robustness check, we repeat our main analyses using the upper and lower bound approach in the Appendix, obtaining very similar results.

⁵See Table IA.1. The correlation of the reuse rate obtained using the proportional measure with the reuse rate from both the upper and lower bound is very high at 0.98. The upper and lower measure also have a correlation of 0.93.

4 How collateral reuse responds to changes in safe asset scarcity

In this section, we empirically analyze how banks respond to changes in collateral scarcity. We first investigate how they adjust their overall reuse of collateral when it becomes more scarce as a result of central bank purchases. Moreover, we distinguish between the intensive and extensive margins of this adjustment, captured by changes in the reuse rate and the amount of incoming collateral.

4.1 The effects of asset scarcity on collateral reuse

We start by analyzing how market participants adjust their reuse of collateral to a shock in collateral supply. We exploit the Eurosystem’s purchases of government bonds via its Public Sector Purchase Programme (PSPP) as a measure of variation in safe asset shortage. The PSPP was announced on 22 January 2015 and consists of the large-scale purchase of bonds issued by euro area governments, agencies and European institutions. The program started on 9 March 2015 and is restricted to purchases in the secondary market. The majority of securities bought under the program are acquired by the national central banks. The geographic allocation of PSPP purchases closely tracks the national central banks’ subscription to the ECB capital key. By the end of our sample period in December 2017, total PSPP purchases reached almost EUR 1.9 Tn.⁶ In our analysis, we make use of proprietary security-level information on Eurosystem PSPP purchases.

Our approach follows that of the literature studying the effects of central banks’ asset purchases on bond yields (De Santis and Holm-Hadulla, 2020; Schlepper, Hofer, Riordan and Schrimpf, 2020) or on bond specialness (D’Amico et al., 2018; Arrata et al., 2020; Corradin and Maddaloni, 2020). Our baseline panel regression specification is the following:

$$\begin{aligned}\Delta \log(\text{Reuse})_{i,t} = & \beta_0 + \beta_1 \text{Asset Purchase}_{i,t} + \gamma' \text{Controls}_{i,t} \\ & + \alpha_i + \alpha_{m,t} + \varepsilon_{i,t},\end{aligned}\tag{3}$$

⁶See <https://www.ecb.europa.eu/mopo/implement/omt/html/index.en.html>

where $\Delta \log(\text{Reuse})_{i,t}$ is the change in the (logarithmic) amount of reused collateral of bond i over month t . To calculate reuse at the bond-level, we aggregate the total amount of that security reused across all banks. The main explanatory variable of interest is $\text{Asset Purchase}_{i,t}$, the amount of bond i that is purchased in the same month by the Eurosystem, measured in percent of the total amount outstanding. If market participants expand their collateral reuse in response to a tightening of supply, we expect a positive sign for the coefficient β_1 .

In addition, we control for various factors that capture changes in the supply and demand of specific collateral. A bond’s supply to the repo market increases if that particular bond reopened for auction, thus raising its total amount outstanding. We therefore control for changes in the amount outstanding. In the government bond market the most recently issued bond of its type (“on the run”) is often more liquid than the previously issued bond (“off-the-run”) (Krishnamurthy, 2002). That said, reissuances are common for Bunds and therefore on-the-run effects are much less important than for U.S. Treasuries (Ejsing and Sihvonen, 2009). Still, since on-the-run bonds are often sought after on the repo market (Jordan and Jordan, 1997) we control for the on-the-run status using a dummy variable. Another reason for a bond to be in high demand is when it becomes the cheapest-to-deliver in the futures market (Buraschi and Menini, 2002; Ejsing and Sihvonen, 2009; Brand et al., 2019). Some investors will have difficulties buying bonds that they need for futures delivery. To avoid penalties from a failure to deliver, these investors will borrow it in the repo market, leading to a high demand for this bond. We therefore also control for the cheapest-to-deliver status. Bond fixed effects α_i capture any security-specific variation. Following Arrata et al. (2020), we also include maturity bucket $\times$ time fixed effects $\alpha_{m,t}$ to account for effects related to the issuer (e.g. rating changes), the yield curve (e.g. haircuts) and market-wide variation. As in Arrata et al. (2020), we define maturity buckets for one to two years, two to five years, five to ten years, and ten to thirty years. Standard errors are clustered at the bond level.

In addition to including these granular fixed effects, we control for two confounding factors that may have affected the supply and demand of specific Bund securities. First, we include information on the utilization of the Bundesbank’s securities lending facility at the bond level. This lending facility was introduced shortly after the beginning of the PSPP with the aim to alleviate the scarcity of specific Bund collateral. Indeed, a number of studies have shown that securities lending facilities mitigate scarcity in the repo market (Baltzer, Schlepper and Speck, 2022; de Souza and Hudepohl, 2024; Greppmair and Jank, 2023; Pelizzon et al., 2024). Initially, the lending was carried out as combined repo/reverse repo transactions. Hence, a participating bank could only borrow a Bund security from the Bundesbank in exchange for an equivalent bond as collateral. In December 2016, the Bundesbank then adapted its procedure and started making securities available against cash collateral. Here, we control for the log change of the securities lending facility (SLF) utilization of a specific bond both against other securities as well as against cash.

We also control for security-specific variation related to the reissuance cycle. The German federal debt management office (Bundesfinanzagentur) frequently reissues Bund securities to maintain liquidity and market stability. This practice ensures a continuous and predictable supply of government bonds and may have an effect on the reuse activity of German banks.⁷ To account for the reissuance cycle, we include a dummy taking the value of one if the last day of the month (the cutoff-day for our monthly reuse data) falls into the five-day window preceding a new reissuance of the specific security, and another one taking the value of one if the last day of the month falls into the five-day window right after a new reissuance.

Specification (1) in Table 2 shows the results from this baseline panel regression. The coefficient for *Purchases* is positive and statistically significant at the 5% level. Hence, banks respond to collateral scarcity by increasing the rate at which they reuse available

⁷Figure IA.1 in the Internet Appendix displays the repo rate dynamics of German sovereign bonds around reissuance events. Similar to evidence for U.S. Treasuries discussed in Keane (1995), the chart documents that repo rates are significantly lower in the days before reissuances and higher afterwards.

collateral. The coefficient estimate shows that a one percentage point purchase of the Eurosystem as a share of the total outstanding of a bond increases the amount of collateral reused by 0.13%.

Banks may adjust their collateral reuse through different channels. On the one hand, they can increase the rate at which they reuse received collateral. We refer to this as the intensive margin of collateral reuse. On the other hand, dealers can seek to borrow more collateral from other market participants in order to reuse it in further transactions, see Equation (2). We refer to this second channel as the extensive margin of collateral reuse. To study the effect of scarcity on the intensive and extensive margins of collateral reuse, we repeat the above regression for $\Delta Reuse\ rate$ and $\Delta \log(Incoming)$ as dependent variables. To calculate the reuse rate at the bond level, we aggregate the total amount of collateral reused across all banks and divide it by the total amount of incoming collateral, also aggregated across all banks.

Columns (2) and (3) of Table 2 report the results of these regressions. At the intensive margin, the rate at which collateral is reused increases by 1.54 percentage points for a purchase of 1% of the amount outstanding, an effect which is highly statistically significant. At the extensive margin, the volume of incoming collateral increases by 0.10% for the same purchase. Although this coefficient narrowly misses statistical significance at the 10% level, we can conclude that banks actively adjust to collateral scarcity along both margins. They reuse existing collateral more efficiently and they also slightly increase their borrowing of a specific piece of collateral when the bond becomes more scarce.

Dealer-bond-level analysis. One might be worried that these bond-level results are driven by large dealer banks. To address this concern, Columns (4)-(6) in Table 2 leverage the granularity of our data and repeat the same analysis at the dealer-bond level. The coefficient estimate for the amount reused in Column (4) is even higher compared to the bond level analysis at 0.21, and equally significant at the 5% level. The coefficient

estimates for $\Delta Reuse\ rate$ and $\Delta \log(Incoming)$ as dependent variables in Columns (5) and (6) are of a similar magnitude as in the bond-level regressions.

Instrumental variable analysis. Our baseline analysis relies on identification within securities: differential PSPP purchases lead to an increased reuse of specific securities. A rich set of security-specific controls and fixed effects absorb any other demand and supply factors and structural drivers that might have also affected collateral markets in our sample period.

To tease out even more precisely the effects of scarcity induced by asset purchases on collateral reuse, we conduct an instrumental variable analysis. This makes use of several design restrictions on the PSPP purchases which significantly affected the universe of bonds, specifically German Bunds, that was eligible for purchase at different points in time. Specifically, we construct a variable that counts the number of days that a bond was eligible for purchase under the PSPP as a fraction of all days since the inception of the program. The details are provided in Appendix B. We show that this variable is a strong instrument for our PSPP share regressor, and that the exogenous component of PSPP purchases is associated with highly significant and sizable increases in all three measures of reuse activity as in our baseline analysis. Overall, the results in this section confirm that scarcity induced by Eurosystem asset purchases induced banks to increase their reuse of collateral at both the intensive and extensive margin.

4.2 Differences in credit quality

The previous analysis shows that asset purchases significantly increase the reuse of German sovereign bonds as collateral. A natural question is whether this result extends to other safe assets that the ECB purchased. We address this question by repeating the above regression at the bond and dealer-bond level for a broader set of euro area sovereign debt. Specifically, we consider sovereign bonds issued by euro area countries with a remaining

maturity between 1 and 30 years and denominated in Euro. Furthermore, in line with the PSPP eligibility criteria we require an investment grade rating (BBB or higher) for at least one of the three major rating agencies and restrict our analysis to countries for which BrokerTec provides repo rate information.⁸ We apply the same filters as described in Section 3 to our data before calculating our reuse measures.

Figure 2 presents an overview of the extended sample of euro area bonds. The upper panel displays the aggregate market value of reused collateral over time, distinguishing between domestic (i.e., German) sovereign bonds and those issued by other euro area countries. The figure highlights that the vast majority of collateral in our sample is domestic. The lower panel shows the average share of reused collateral by issuer country and by credit rating. German bonds constitute the largest share of collateral, followed by Italian and French sovereign bonds. With respect to credit ratings, 65.4% of reused collateral is AAA-rated, 20.9% is AA-rated, and 13.8% is rated A or BBB.

When studying the effect of asset purchases on reuse, we consider different sample splits. First, we differentiate between German Bunds and all non-German sovereign debt that was purchased under the PSPP (and as such had a minimum credit rating of BBB) and was held as collateral by the German banks in our sample. Second, we differentiate between collateral with credit ratings of AAA versus non-AAA. Lastly, we investigate AAA-rated bonds excluding German collateral. An important caveat to the analysis of the extended sample is that it includes only German banks. Thus, home bias may affect collateral selection, potentially skewing the sample towards domestic securities.

The results are provided in Table 3. Panel A provides the analysis at the bond level. We first distinguish between German and non-German sovereign debt as collateral. Note that we do not have access to exactly the same control variables in this European sample, and thus the results for Germany differ slightly with respect to our baseline analysis in

⁸The BrokerTec data covers all major euro area sovereign debt markets. Specifically, our analysis includes sovereign bonds issued by Austria, Belgium, Germany, Spain, Finland, France, Greece, Ireland, Italy, Netherlands, and Portugal.

Table 2. That said, the results confirm that for German Bunds all three measures of reuse activity – the monthly growth rate of reuse, the change in the reuse rate, as well as the growth rate of incoming collateral – respond positively and significantly to asset purchases. However, the same is not true for non-German sovereign debt. Only overall reuse is significant in the corresponding regression.

A similar result is obtained when we apply a slightly different sample split, separating AAA-rated from non-AAA rated euro area collateral. In our sample, in addition to German Bunds only sovereign debt of Finland, Austria and the Netherlands had the highest achievable credit rating for at least one of the three major rating agencies. By including these countries, we gain about one thousand observations relative to the German sample. Including these securities, the results are very similar to those for Bund collateral both in terms of estimates as well as significance. Lastly, we focus on AAA-rated bonds excluding those issued by Germany. For these safe assets, we find no significant effects, most likely due to a lack of power in this small sample.

We therefore repeat our analysis using the dealer-bond sample in Panel B of Table 3. This significantly increases the sample size and allows for more precise estimation by including dealer-time fixed effects. For German bonds and AAA-rated bonds, we find significant effects of asset purchases on our reuse measures. In contrast, for non-German bonds and non-AAA-rated bonds, the effect is significant in only one specification. For non-German AAA-rated bonds, the reuse rate increases by a similar magnitude as for German Bunds, but the coefficient is only significant at the 10% level.

Overall, these results highlight that asset purchases affect reuse significantly, but that this effect is concentrated in the highest quality euro area safe assets, predominantly German Bunds.

4.3 Extensions

In this section we discuss several extensions of the previous analysis, which we describe in more detail in the Appendix.

The granularity of our data allows us to study whether the sensitivity of reuse to scarcity depends on the type of collateral. In Appendix C, we conduct two analyses to shed light on this question. First, we assess whether on-the-run bonds are more sensitive to asset purchases than off-the-run bonds. Our results provide some evidence that this is the case for European sovereign bonds. Second, we distinguish between collateral with different remaining time to maturity. Specifically, we repeat the sensitivity analysis of the reuse rate for subsets of shorter-dated collateral (bonds with a remaining maturity between 1 and 5 years, medium-dated collateral (remaining maturity between 5 and 10 years) and longer-dated collateral (remaining maturity between 10 and 30 years). For both German and European collateral we observe a decreasing “term-structure” of sensitivities to asset purchases: the reuse of collateral with shorter and intermediate remaining time to maturity is more sensitive to asset purchases compared to longer-term collateral.

Finally, in Appendix D we provide a back-of-the-envelope calculation to put the estimated increase of collateral reuse in response to asset purchases into perspective. We ask the following question: Given a specific reduction in the supply of collateral, how much additional collateral do banks need to provide through collateral reuse in order to maintain a constant amount of collateral in the market? We find that our estimated coefficients are considerably higher than this calculation would suggest. This apparent overcompensation is likely due to central banks purchasing disproportionately from holders that would otherwise supply these assets as collateral. Consistent with this interpretation, [Koijsen, Koulischer, Nguyen and Yogo \(2021\)](#) find that euro area banks were an important net seller of euro area government bonds to the Eurosystem in the PSPP, second only to foreign investors.

5 Collateral reuse and repo market specialness

Specialness as a measure of safe asset scarcity depends on the demand for and supply of the specific collateral, but is also a function of the transaction costs faced by investors owning the collateral (Duffie, 1996). In the previous section, we have shown that banks respond to safe asset scarcity induced by central bank asset purchases by adjusting their reuse activity. Hence, they effectively increase the supply of specific collateral in the repo market. However, there is a binding constraint as dealers cannot post more collateral than they receive from other market participants or own outright. As a result, a bond’s scarcity in the repo market may depend on its prevailing level of reuse. In this section, we therefore explore how the specialness of specific collateral depends on banks’ reuse activity.

In Duffie (1996) and subsequent work, specialness spreads are commonly computed by subtracting from a bond’s specific collateral rate the general collateral rate as a proxy for the risk-free funding rate. However, as Arrata et al. (2020) point out, the general collateral rate may be a biased benchmark when all eligible bonds are on special. We follow their approach and use the specific collateral rate itself instead of the specialness spread to measure bond scarcity premia. Any changes in the general collateral rate will be absorbed by the time fixed effects included in our regressions.

To evaluate the effect of collateral reuse on scarcity premia, we regress changes in the specific collateral repo rate $\Delta repo\ rate_{i,t}$ on our measures of reuse activity and a set of controls, including the lagged change in repo rates:⁹

$$\begin{aligned} \Delta repo\ rate_{i,t} = & \beta_0 + \beta_1 Asset\ Purchase_{i,t} + \beta_2 Reuse_{i,t-1} + \beta_3 Asset\ Purchase_{i,t} \times Reuse_{i,t-1} \\ & + \gamma' Controls_{i,t} + \alpha_i + \alpha_{m,t} + \varepsilon_{i,t} \end{aligned} \quad (4)$$

⁹In a dynamic panel model estimates can be biased, but this bias is decreasing with the length of the sample T (Nickell, 1981). Given the sample length of $T = 34$ in our application, the Nickell bias is negligible irrespective of the persistence of the endogenous variable (Phillips and Sul, 2007).

To account for the potential endogeneity between reuse and repo rates we interact PSPP purchases with the *lagged* level of collateral reuse. The intuition is as follows: it should be easier for market participants to expand their reuse activity in bonds with a low prevailing level of reuse. As a result, repo rates should be less sensitive to purchase-induced scarcity. In contrast, dealers may not be able to compensate the reduced supply for bonds that are already heavily reused. We thus expect a negative coefficient on this interaction term. We again consider the effects of reuse at the extensive and the intensive margin, captured by the overall amount of collateral reused and the reuse rate.

For comparability, both measures of reuse are demeaned and standardized to have unit variance. *Controls* account for lagged changes in the repo rate and, as above, changes in the amount issued and on-the-run and cheapest-to-deliver status. Moreover, we again include the security-level take-up of the Bundesbank’s securities lending facility, both against other securities and against cash as collateral. We further control for the reissuance cycle by including a dummy taking the value of one if the last day of the month falls into the five-day window before or after a reissuance. In addition, we include bond fixed effects and maturity bucket×time fixed effects. Standard errors are clustered at the bond level.

Table 4 presents the estimation results. In Column (1) we estimate the baseline regression without the interaction term. Consistent with the prior literature, we find that asset purchases compress repo rates, i.e. they increase bonds’ specialness. An asset purchase of one percent of the amount outstanding reduces the bond’s repo rate by a little less than one basis point. This is similar in magnitude to [Arrata et al. \(2020\)](#), who report a reduction of 0.78 basis points with regard to a one percent PSPP purchase using daily data for a broader cross-section of euro area sovereign bonds. Note that this effect is over and above the impact of the lagged repo rate which features a highly statistically significant negative coefficient, indicating a substantial amount of mean reversion in repo rates. Importantly, the impact of asset purchases on repo rates is also above and beyond the impact of the utilization of the Bundesbank’s securities lending facility against cash

collateral. The negative coefficient on this control variable indicates that a higher uptake of the SLF is associated with more negative repo rates. However, this variable should be interpreted with caution, as it primarily reflects increased SLF usage in response to scarcity. Unlike this coefficient, [Greppmair and Jank \(2023\)](#) show that increased utilization of the SLF – driven by a quasi-natural experiment – alleviates scarcity in the repo market.

In Columns (2) and (3) we include the two measures of reuse as well as the interaction term with PSPP purchases. The coefficients on lagged $\log(\textit{Reuse})$ and the lagged $\log(\textit{Reuse Rate})$ are both positive and highly statistically significant. Hence, bonds that are reused more strongly are associated with higher repo rates and are thus less special. Crucially, for both measures of reuse the interaction term is significant, both statistically and economically. A one standard deviation increase of reuse in Column (2) further lowers the special repo rate by half a basis point for each percent share of amount outstanding purchased. This corresponds to an increase by more than one half with respect to the baseline sensitivity of repo rates to Eurosystem asset purchases. The effect for a similar increase in the reuse rate, shown in Column (3), is even stronger, implying an additional reduction in the repo rate by 0.69 basis points for each percent of amount outstanding purchased. Hence, when the prevailing amount or rate of reuse of a bond is high, the scarcity induced by central bank asset purchases increases the bond’s specialness.

Table [IA.7](#) in the Internet Appendix repeats this analysis for a larger set of euro area bonds for which the banks in our sample are comparably active, yielding very similar results. Hence, our finding that reuse affects the degree to which asset purchases increase specialness is not specific to German Bunds.

How persistent are the effects of central bank purchases on the specialness of specific collateral? And does the prevailing degree of reuse affect the persistence? While we only observe reuse monthly, we can study the dynamic effects of asset purchases on specialness

premiums at the daily frequency. To this effect, we estimate panel local projections in the spirit of Jordà (2005). Specifically, for each horizon h , we run the following regression

$$\Delta repo\ rate_{i,t+h} = \beta_{0,h} + \beta_{1,h} Asset\ Purchase_{i,t} + \beta_2' Controls + \alpha_i + \alpha_t + \varepsilon_{i,t+h} \quad (5)$$

and collect the coefficients $\beta_{1,h}$. These measure the dynamic effects of an asset purchase of 1% of the amount outstanding on the specialness premium h days out. Controls in the regression include the log of amount outstanding, a dummy indicating on-the-run status, a dummy indicating cheapest-to-deliver status and α_i and α_t indicate bond and time fixed effects, respectively.

To assess the impact of collateral reuse on the transmission of asset purchases to scarcity premiums, we perform a split-sample analysis. Specifically, we run the above regression separately for bonds for which the reuse rate is below or above the 25th percentile, which corresponds to a reuse rate of 63%. The results are provided in the top panels of Figure 3. They show that central bank asset purchases have fairly persistent effects on repo rates, and that these effects are stronger and last longer for bonds which feature a higher reuse activity. Specifically, the repo rate drops by about 10 basis points for securities featuring a high reuse rate in the first few days after the purchase, and the repo rate remains compressed at that lower level for at least a month. In contrast, the dynamic impact of purchases is less strong and less persistent for Bund securities that feature a low reuse rate.

The bottom panel of Figure 3 plots the difference in coefficients $\beta_{1,h}$ across the two groups of bonds for each horizon h along with the 90% confidence interval. While there is little difference in the effect of purchases on specialness premiums in the first few days, the wedge widens substantially over time and is statistically significantly different from zero after 20 trading days. Hence, asset purchases increase specialness premiums more strongly

and substantially more persistently for bonds which feature a high prevailing degree of reuse activity.

Overall, the results in this section highlight the importance of collateral reuse in compensating asset scarcity. Repo rates respond more strongly and persistently to scarcity induced by asset purchases for bonds where reuse activity is already high.

6 Collateral reuse and market functioning

In the previous section, we have shown that banks can dampen scarcity effects by reusing available collateral more efficiently. However, specialness premia are more pronounced and increase more persistently in response to central bank purchases for securities with elevated reuse activity.

In addition to this asymmetry, the reuse of collateral may also increase the interconnectedness among market participants and thereby propagate or even amplify shocks ([Brumm et al., 2023](#)). As a result, market functioning may be impaired. In this section, we analyze the degree to which an increased reuse activity affects the functioning of the repo market. We first study the link between reuse rates and delivery failures and volatility in the repo market. We then quantify the impact of reuse on the effect of asset purchases on mispricing in the secondary bond market.

6.1 Collateral reuse and repo market functioning

The repo market is an essential building block of the financial system, enabling participants to meet short-term funding needs and to obtain specific securities for other transactions. The smooth functioning of the repo market thus ensures efficient capital flows and supports market stability. In this section, we study the role of collateral reuse in shaping the effects of purchase-induced scarcity on repo market quality.

We measure the effective functioning of the repo market using three proxies capturing the prevalence and duration of delivery failures and repo rate volatility. Fails-to-deliver

indicate instances where a party does not fulfill its obligation to deliver the agreed-upon security in a repo transaction, reflecting potential counterparty risk and operational inefficiencies. Monitoring fails is essential for gauging repo market conditions and the reliability of counterparties.¹⁰ On the other hand, repo rate volatility, or fluctuations in the interest rates in the repo market, provides insights into the overall liquidity and stability of the financial system. Sudden spikes or fluctuations in repo rates can signal stress in the market, impact borrowing costs and affect broader financial conditions.

We obtain security-level information on failures to deliver for Bund securities from LCH Limited, which is the clearinghouse for various European repo trading platforms, including Euro-MTS, MTS-France, MTS Italy, MTS Associated Markets, CME (BrokerTec), TP Repo (TP ICAP) and Tradeweb. Based on the daily fails data, we compute two measures of repo market dysfunction for the monthly frequency at which we observe reuse. First, we compute the security-specific fails rate in a given month as the share of number of days in the month when the security failed to deliver. Second, to capture the persistence of delivery failures, we construct a dummy variable that takes on the value of one when a security experienced at least five consecutive days of delivery failures in a month.¹¹ Finally, we measure repo rate volatility as the logarithm of the standard deviation of daily repo rates over the course of a month.

Figure 4 provides binned scatter plots of bonds' fails rate (top panel) and repo-market volatility (bottom panel) against their lagged reuse rate. We differentiate between bonds

¹⁰Garbade et al. (2010) argue that if security lenders observe a growing incidence of settlement fails, they may cut down on lending out of concern that their securities may not be delivered on a timely basis. Hence, an increased prevalence of fails might increase scarcity of specific collateral and thus lead to fragility in funding markets. In line with this, Fleming and Garbade (2005) document that in the aftermath of September 11, 2001, some securities lenders withdrew from the repo market which led to a rise in special repo rates and perpetuated "daisy chains" of fails. Finally, as shown by Iyer and Macchiavelli (2017), dealers failing to receive securities are highly likely to also fail to deliver the same securities, suggesting a high degree of rehypothecation of collateral and matched-book activities of dealer banks. Hence, chains of fails result in a market-wide deterioration of liquidity.

¹¹The literature commonly refers to "daisy chains" of cascading fails as a situation when multiple counterparties fail to deliver a specific security within the same period (e.g., Fleming and Garbade, 2005). This is related to but different from the persistence of aggregate fails over multiple periods (Garbade et al., 2010), which we can measure with the available data.

that trade on special (specialness spread greater or equal to 5 bps) and those that don't (specialness spread below 5 bps). We group variables into equal-sized bins along the x-axis.

The upper chart shows that securities which are on special experience higher fails rates. More importantly, however, repo fails are strongly positively associated with the lagged reuse rate of a bond for both special and non-special securities. For bonds that trade on special, this relationship is strongly nonlinear. In particular, those with elevated reuse rates experience considerably higher delivery failures in the following month.

The bottom panel of Figure 4 shows a similar pattern in the relation between repo rate volatility and lagged collateral reuse. Securities that trade on special feature substantially more volatile repo rates. While the repo rate volatility increases in the previous month's reuse rate, the relationship is again visibly convex for special bonds. While reuse rates below 60% do not seem to be associated with increased subsequent volatility, the slope is strongly positive for reuse rates above that level. Combined, these charts thus provide initial evidence that excessive reuse of collateral may be associated with impaired repo market functioning reflected by delivery failures and heightened rate volatility.

Of course, other factors related to the supply and demand of collateral may affect both their reuse and delivery fails and repo rate volatility. We account for such potential confounding factors in regressions of the repo market functioning indicators on lagged collateral reuse. Specifically, we control for the lagged dependent variable and a comprehensive set of security-specific controls as well as time fixed effects. The regression thus explores the cross-sectional relation between collateral reuse and delivery fails in the next month. For repo market volatility we proceed likewise, using the logarithm of repo rate volatility as dependent variable.

Panel A of Table 5 reports the results for the fails rate. The first column shows that there is a positive relation between collateral reuse and fails in the next month. A one standard deviation higher reuse rate (23 percentage points) raises the fails rate by $3.25 \times 23 \approx 75$ basis points. In Column (2), we add a dummy for bonds that trade on

special, which we again define as securities with a specialness spread above 5 basis points. While the coefficient indicates that special bonds on average experience a 50 basis points higher fails rate in the next month, controlling for the lagged reuse rate the coefficient is not statistically significant. In the third column, we add an interaction term of the lagged reuse rate and the specialness dummy. This interaction term is highly statistically significant while the coefficient for reuse itself becomes insignificant. This implies that reuse is only associated with a higher fails rate when the bond is special and difficult to borrow in the market for collateral.

In Panel B, we use as dependent variable a dummy that equals one if a security has failed to deliver on at least five consecutive days in a month. The results are consistent with those for the average fails rate. A one standard deviation higher reuse rate is associated with a two percentage point ($0.095 \times 0.23 \approx 0.02$) higher chance of such spells of delivery failures. The subsequent columns document that this effect is entirely driven by bonds that trade on special *and* feature a high reuse rate. Column (6) shows that the probability of extended spells of consecutive fails increases by an economically significant ten percentage points ($0.44 \times 0.23 \approx 0.1$) for a one standard deviation increase in the reuse rate when the bond is special and the reuse rate is already above its 25th percentile.

Panel C of Table 5 repeats the previous analysis for the monthly repo rate volatility as dependent variable. Consistent with our findings from Figure 4, there is a positive and convex relationship between collateral reuse and repo market volatility in the next month. Bonds with high reuse rates feature more volatile repo rates in the coming month. This is true also for securities that trade on special. But it is particularly bonds that are special *and* at the same time feature a high prior reuse rate which experience a higher subsequent repo rate volatility. As the split-analysis shows, this effect is exclusively driven by securities with a reuse rate above 60%.

In sum, these results show that when securities that are special are reused more intensively, the probability of repo market disruptions sharply increases. This highlights

the tradeoffs associated with collateral reuse. While reuse can help to absorb shocks by making more collateral available, high levels of reuse increase the interconnectedness of market participants and can thus lead to an amplification of shocks in the repo market.

6.2 Collateral reuse and mispricing in the cash bond market

A previous literature has established a close connection between specialness in the repo market and pricing conditions in secondary government bond markets (Duffie, 1996; Jordan and Jordan, 1997). Moreover, it has been shown that purchase-induced bond scarcity can result in mispricing in the secondary market for government bonds (D’Amico et al., 2018). In light of these results, this section investigates the role of collateral reuse for the link between repo market specialness and pricing in the cash market.

To measure the degree of mispricing, we follow Hu, Pan and Wang (2013). Specifically, we calculate the yield fitting error for a given security as the difference between the par yield and the yield implied by the curve fitting algorithm popularized by Nelson and Siegel (1987) and later extended by Svensson (1994). Denoting this yield fitting error difference $\Delta Y_{i,t}$, we run the following regression:

$$\begin{aligned}\Delta Y_{i,t} = & \beta_0 + \beta_1 \Delta Specialness_{i,t} + \beta_2 Asset\ Purchase_{i,t} \\ & + \beta_3 \Delta Specialness_{i,t} \times Asset\ Purchase_{i,t} \\ & + \gamma' Controls_{i,t} + \alpha_i + \alpha_t + \varepsilon_{i,t}.\end{aligned}\tag{6}$$

Table 6 summarizes the results. The first column only uses the daily change in the specialness spread as explanatory variable. The statistically significant coefficient shows that a one percentage point increase in the specialness spread lowers the yield in the secondary market by 10 basis points compared to the Nelson-Siegel-Svensson curve. Hence, in line with the findings by Jordan and Jordan (1997) for U.S. Treasuries, German Bunds which are more special in the repo market also are more expensive in the secondary market.

The second column adds the PSPP purchase relative to the total amount outstanding and an interaction term with specialness as explanatory variables. The coefficients on both terms are significantly negative, indicating that central bank purchases further compress the secondary market yield of government bonds beyond the effect of specialness. The negative coefficient on the interaction term shows that a one percent higher purchased amount lowers the yield by an additional 18 basis points for the same degree of specialness. This finding is in line with [D’Amico et al. \(2018\)](#) who use a purchase dummy to document that on days of Federal Reserve asset purchases, Treasury yields in the secondary market are significantly lower than on other days.

The last two columns repeat the same regression for different sample splits that allow us to study the role of collateral reuse for the link between repo specialness and secondary cash market pricing. Column (3) shows the results for securities with low reuse rates in the prior month. While specialness has a larger impact on the yield fitting error in this subsample, there is no additional impact from Eurosystem purchases. This shows that when reuse is low, banks can compensate the scarcity induced by asset purchases and cushion the spillover to the cash bond market. The results are very different for the sample of securities with high reuse rates in the previous month. Here, the direct impact of specialness on yield fitting errors is negligible. That said, purchases significantly amplify the effect of specialness on secondary market yields. This finding is intuitive based on our previous results: when reuse is already high, banks cannot increase their reuse in response to central bank purchases sufficiently to mitigate their effects on pricing conditions in the secondary market.

In sum, the findings in this section highlight that banks’ collateral reuse in response to purchase-induced scarcity involves a clear tradeoff. When reuse is low, banks can easily adjust it and, as a result, mitigate the effects of central bank purchases on market functioning. When reuse is already high, however, the impact of asset purchases on market functioning gets amplified.

7 Conclusion

In this paper we document that banks adjust to safe asset scarcity by making more efficient use of received collateral. In response to sovereign debt purchases by the Eurosystem, dealers increase their collateral reuse activity. The increase in collateral reuse absorbs part of the supply reduction, which is reflected in a lower scarcity premium on the repo market following an asset purchase. But increasing collateral reuse also has a downside: high levels of collateral reuse are associated with higher rates of delivery failures and increased volatility in the repo market.

Our results suggest that theories of repo market specialness should account for collateral reuse. While existing models such as [Duffie \(1996\)](#) relate specialness to supply frictions, there is no explicit role for reuse to affect the degree of these frictions. Specifically, our findings highlight that the supply of specific collateral is endogenous as dealers adjust their reuse to changing market conditions. However, they are constrained in doing so as they cannot post more collateral than they own outright or receive from other market participants. This constraint, in turn, depends on the prevailing level of reuse and may thus be occasionally binding. Hence, the effects of exogenous changes to the supply of safe assets are potentially time-varying and nonlinear.

From a policy perspective our results highlight a new tradeoff between unconventional monetary policy and financial stability. As a side effect to quantitative easing, asset purchases increase collateral reuse. High reuse rates, in turn, are associated with lower repo market quality. More generally, our results suggest that global supply and demand imbalances for safe assets impact financial markets beyond the resulting premia for safe assets. Market participants' endogenous response to safe asset scarcity through collateral reuse may ultimately result in an amplification of shocks.

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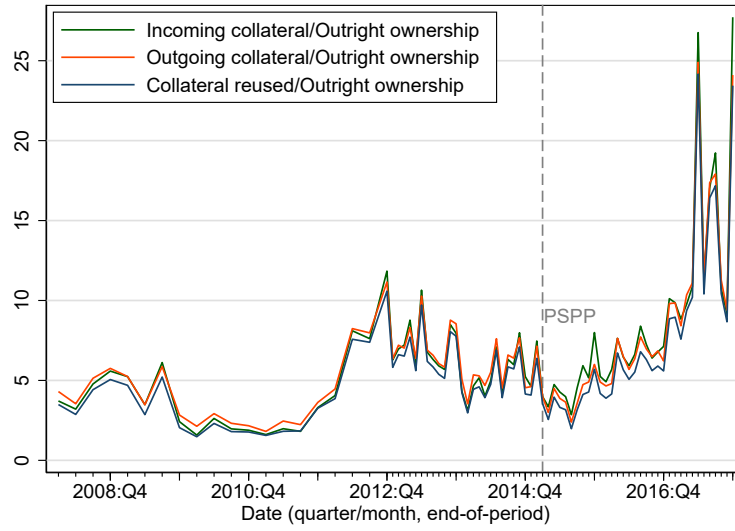
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(a) Collateral over Outright Ownership



(b) Reuse Rate

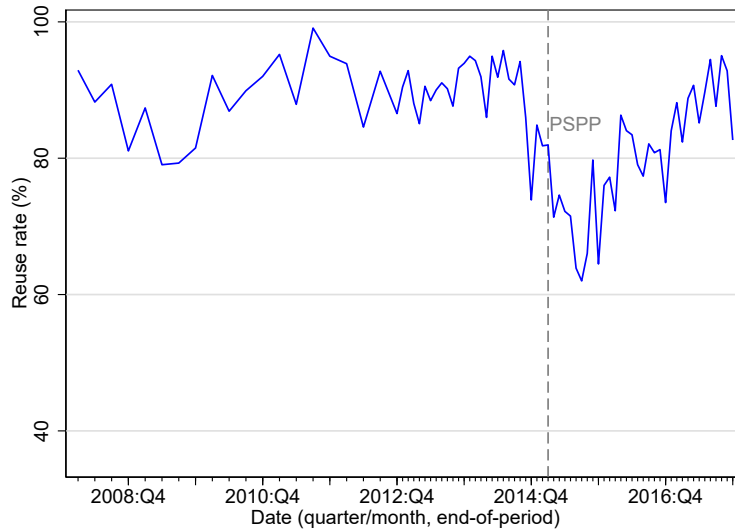
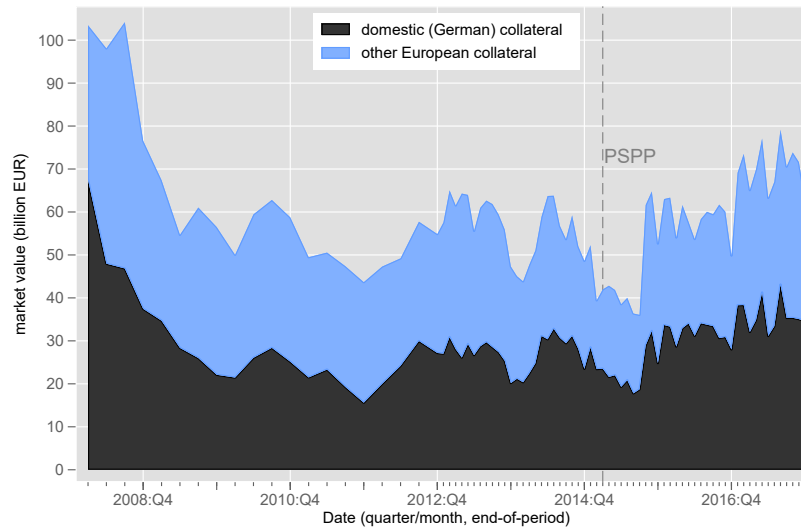
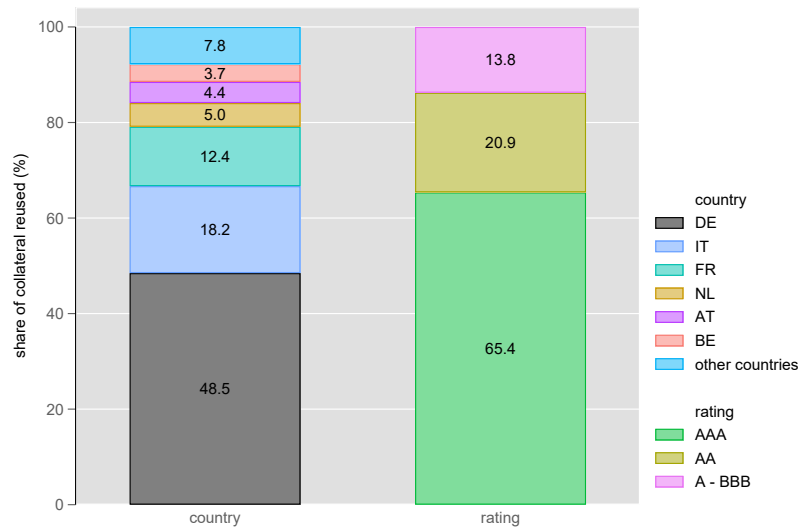


Figure 1: Collateral reuse over time

This figure shows the development of aggregate collateral reuse for German sovereign bonds with remaining maturity between 1 and 30 years. The upper panel plots the multiplier obtained by dividing the amount of incoming, outgoing, and reused collateral in German sovereign bonds by the amount of bonds owned outright. The lower panel shows the development of the aggregate collateral reuse rate. The sample period is 2008-2017, 2008-2010 at quarterly frequency, 2013-2017 at monthly frequency.



(a) Collateral Reused: Domestic vs. Foreign



(b) Collateral Reused by Issuer Country and Rating

Figure 2: Extended Sample: Collateral Reused by Issuer Country and Rating

The top figure shows the market value of collateral reused for domestic (i.e. German) collateral and collateral by other European countries in our sample. The bottom figure shows the overall share of collateral reused, computed as the time-series average, in our sample, by issuer country (left column) and by issuer rating (right column). The group *other countries* includes Spain, Finland, Greece, Ireland, and Portugal. We consider European sovereign bonds with remaining maturity between 1 and 30 years. The sample period is 2008-2017, 2008-2010 at quarterly frequency, 2013-2017 at monthly frequency.

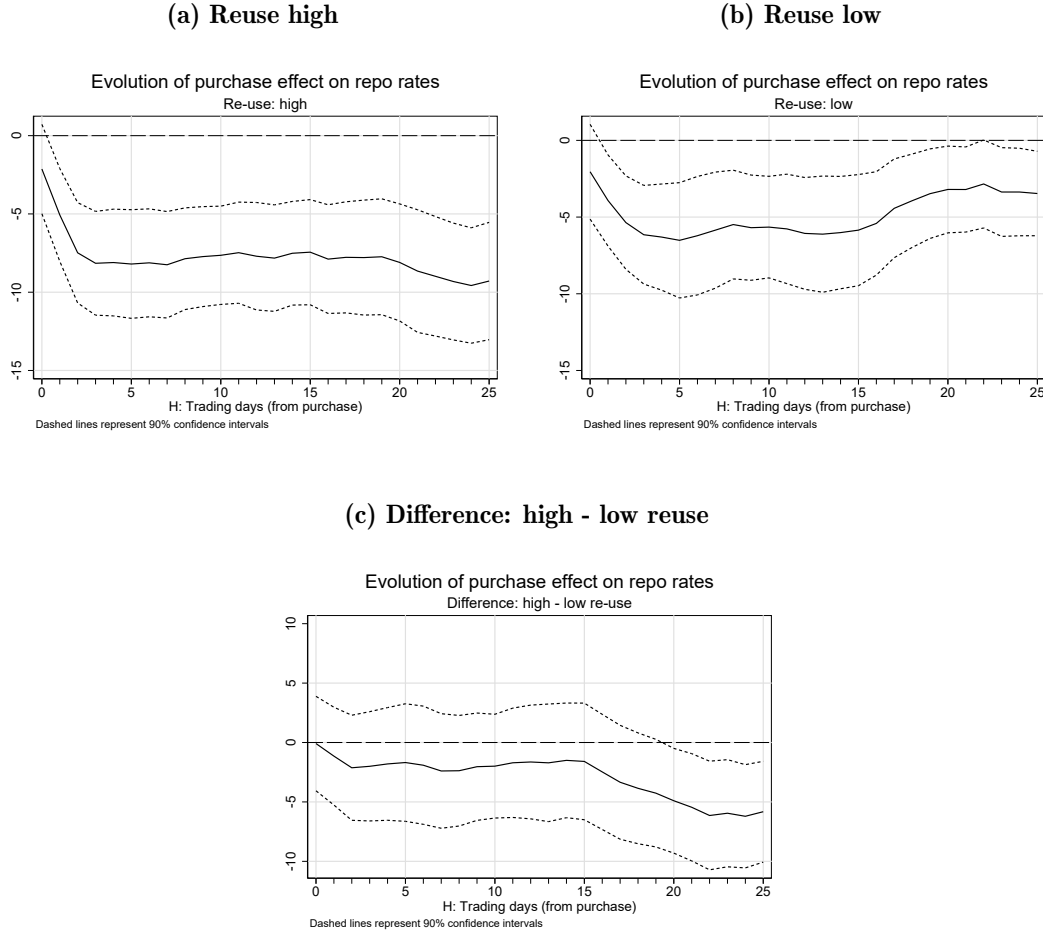


Figure 3: The dynamic effect of asset purchase on repo rates

The figure shows the dynamic effects of asset purchases on repo rates at the daily frequency. We estimate panel local projections in the spirit of [Jordà \(2005\)](#) using dummy interactions, indicating bonds with a high and low reuse rate (the cutoff is the 25th percentile), see Equation (5). The regression controls for the log of amount outstanding, an indicator for on-the-run status, an indicator for cheapest-to-deliver status, dummies for the periods before and after bond reissuance, the log change in the securities lending facility (SLF) separately for cash and securities collateral, as well as time and bond fixed effects. Figure (a) and (b) show the dynamic effects of an asset purchase of 1% of the amount outstanding on repo rates up to 25 days. Figure (c) reports the difference between bonds with a high and low reuse rate. The sample consists of the daily panel of German sovereign bonds with remaining maturity between 1 and 30 years. The sample period is March 2015 - December 2017.

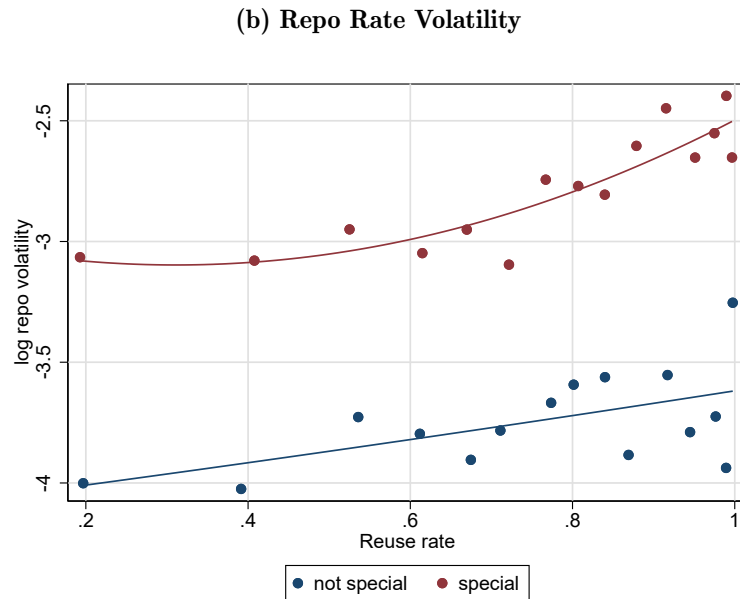
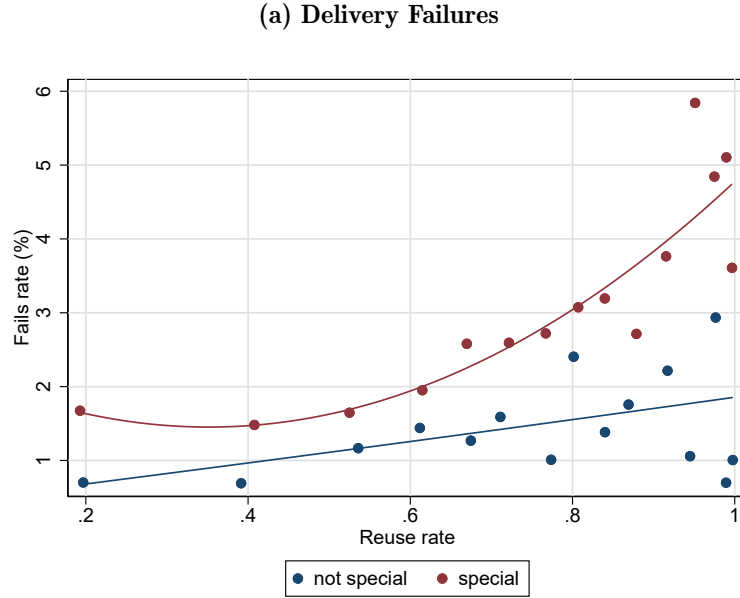


Figure 4: Collateral Reuse and Repo Market Quality

This figure depicts the relationship between reuse rates and measures of repo market quality. The top panel shows the link between reuse rates and the average rate of delivery failures in the next month. The bottom panel provides the relation between reuse rates and repo rate volatility (measured as the logarithm of the standard deviation of repo rates for each month), also in the next month. We differentiate between bonds that are not on special (specialness spread below 5 bps) and bonds that are on special (specialness spread greater or equal to 5 bps). We group variables into equal-sized bins along the x-axis. The sample consists of the monthly panel of German sovereign bonds with remaining maturity between 1 and 30 years. The sample period is March 2015 - December 2017.

Table 1:
Descriptive statistics

This table provides summary statistics of the main variables used in regressions throughout the paper. The sample period is March 2015 - December 2017 at the monthly frequency. Panel A describes the bond-level and Panel B the dealer-bond-level data consisting of German sovereign bonds with a remaining maturity between 1 and 30 years. The variables describe monthly changes in the logarithmic amount of collateral reused ($\Delta \log \text{Reuse}_t$), monthly changes in the reuse rate ($\Delta \text{Reuse Rate}_t$), monthly changes in the logarithmic amount of incoming collateral ($\Delta \log(\text{Incoming})_t$), monthly changes in the repo rate ($\Delta \text{Repo Rate}_t$), monthly average rate of delivery failures (Avg. Fails Rate_t), and repo rate volatility, measured as the logarithm of the standard deviation of daily repo rates for each month ($\log(\text{Repo Rate Volatility})_t$).

Variable	Mean	Std. dev.	Percentiles			N
			25th	50th	75th	
Panel A: bond-time panel						
$\Delta \log(\text{Reuse})_t$	0.05	1.05	-0.28	0.02	0.35	1,694
$\Delta \text{Reuse Rate}_t$	0.22	22.17	-9.65	-0.09	11.35	1,694
$\Delta \log(\text{Incoming})_t$	0.05	0.91	-0.19	0.00	0.23	1,694
$\Delta \text{Repo Rate}_t$	-2.07	29.20	-6.98	-1.47	3.22	1,681
Avg. Fails Rate _t (%)	2.83	4.70	0.13	1.08	3.34	1,714
$\log(\text{Repo Rate Volatility})_t$	-2.99	1.16	-3.81	-3.18	-2.43	1,714
Panel B: dealer-bond-time panel						
$\Delta \log(\text{Reuse})_t$	0.07	8.24	-0.35	0.00	0.39	10,229
$\Delta \text{Reuse Rate}_t$	0.34	46.59	-3.90	0.00	5.29	10,229
$\Delta \log(\text{Incoming})_t$	0.03	6.91	-0.10	0.00	0.09	10,229

Table 2: Asset purchases and collateral reuse: intensive and extensive margin

The table reports the results of a regression of changes in collateral reuse on asset purchases at the monthly frequency. The dependent variable in specifications (1) and (4) is changes in logarithmic amount of collateral reused ($\Delta \log \text{Reuse}_t$). In specifications (2) and (5) the dependent variable is changes in reuse rate ($\Delta \text{Reuse Rate}_t$), and in specifications (3) and (6) changes in the logarithmic amount of incoming collateral ($\Delta \log(\text{Incoming})_t$). The regression controls for changes in the amount outstanding, controls for on-the-run and cheapest-to-deliver status, and includes controls for the log change in the securities lending facility (SLF) separately for cash and securities collateral, as well as for the period before and after bond reissuance using dummy variables. The sample consists of German sovereign bonds with a remaining maturity between 1 and 30 years. Specifications (1) - (3) are based on a bond-time panel, and specifications (4) - (6) on a dealer-bond-time panel. The regression model is outlined in Equation (3). The sample period is March 2015 - December 2017. t -statistics based on clustered standard errors are provided in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	bond level			dealer-bond level		
Dependent variable:	$\Delta \log(\text{Reuse})_t$	$\Delta \text{Reuse Rate}_t$	$\Delta \log(\text{Incoming})_t$	$\Delta \log(\text{Reuse})_t$	$\Delta \text{Reuse Rate}_t$	$\Delta \log(\text{Incoming})_t$
Asset Purchase _t	0.13** (2.00)	1.54*** (2.74)	0.10 (1.65)	0.21** (2.21)	1.27** (2.40)	0.12 (1.34)
Δ Amount outstanding _t	0.01 (1.02)	-0.13 (-1.17)	0.01 (1.59)	0.03 (1.56)	0.05 (0.53)	0.05*** (3.03)
Dummy: On the run _t	0.02 (0.07)	5.71 (1.24)	-0.18 (-0.62)	-0.05 (-0.07)	2.60 (0.66)	-0.53 (-0.76)
Dummy: Cheapest-to-deliver _t	-0.09 (-1.06)	-1.36 (-0.81)	-0.04 (-0.56)	-0.18 (-0.47)	-0.89 (-0.43)	-0.11 (-0.35)
Δ SLF utilization (sec)	-0.01 (-1.22)	-0.04 (-0.32)	-0.00 (-1.08)	0.00 (0.09)	0.02 (0.17)	-0.00 (-0.07)
Δ SLF utilization (cash)	0.00 (0.44)	0.01 (0.05)	0.00 (0.58)	0.01 (0.56)	-0.01 (-0.16)	0.00 (0.29)
Dummy: Before reissuance	-0.14 (-0.20)	3.35 (0.32)	-0.23 (-0.40)	0.24 (0.18)	1.62 (0.21)	-0.15 (-0.11)
Dummy: After reissuance	0.85 (0.88)	-5.00 (-0.65)	1.30 (1.35)	1.40 (1.20)	8.43 (1.40)	0.53 (0.50)
Constant	-0.09 (-1.23)	-1.23* (-1.91)	-0.08 (-1.07)	-0.22* (-1.67)	-1.24* (-1.70)	-0.14 (-1.24)
Fixed effects:						
bond	yes	yes	yes			
maturity bucket $\times$ time	yes	yes	yes	yes	yes	yes
dealer $\times$ time				yes	yes	yes
dealer $\times$ bond				yes	yes	yes
R^2	.2532	.2144	.2924	.1704	.1724	.1138
N	1,639	1,639	1,639	9,644	9,644	9,644

Table 3: Asset purchases and collateral reuse: Differences across credit quality

The table presents the results of a regression analysis examining the relationship between changes in collateral reuse and asset purchases on a monthly basis. Panel A reports results based on a bond-time panel; Panel B reports results based on a dealer-bond-time panel. We differentiate between various definitions of safe assets: (i) German vs. non-German collateral, (ii) AAA-rated vs. non-AAA-rated collateral, and (iii) AAA-rated collateral excluding German assets. In our sample, sovereign debt issued by Germany, Austria, Finland and the Netherlands is rated AAA, while all other euro area countries are rated AA or lower. The dependent variable in specifications (1) and (4) is changes in logarithmic amount of collateral reused ($\Delta \log \text{Reuse}_t$). In specifications (2) and (5) the dependent variable is changes in reuse rate ($\Delta \text{Reuse Rate}_t$), and in specifications (3) and (6) changes in the logarithmic amount of incoming collateral ($\Delta \log(\text{Incoming})_t$). In Panel A, the regression uses a bond-time and in Panel B a dealer-bond-time panel. Both regressions control for changes in the amount outstanding and for on-the-run and cheapest-to-deliver status. Moreover, we include maturity bucket x time, dealer x time, and dealer x bond fixed effects. The sample consists of German and European sovereign bonds with a remaining maturity between 1 and 30 years and a minimum credit rating of BBB. The regression model is outlined in Equation (3). The sample period is March 2015 - December 2017. t -statistics based on clustered standard errors are provided in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable:	$\Delta \log \text{Reuse}_t$	$\Delta \text{Reuse Rate}_t$	$\Delta \log(\text{Incoming})_t$	$\Delta \log \text{Reuse}_t$	$\Delta \text{Reuse Rate}_t$	$\Delta \log(\text{Incoming})_t$
Panel A: Bond level						
	German			Non-German		
Asset Purchase _t	0.13** (2.21)	1.63*** (3.10)	0.10* (1.72)	0.20** (2.04)	0.23 (0.43)	0.06 (1.61)
R^2	.2445	.2114	.2719	.2465	.2733	.2695
N	1,691	1,691	1,691	4,955	4,955	4,955
	AAA			Non AAA		
Asset Purchase _t	0.12** (2.21)	1.46*** (3.17)	0.08* (1.81)	0.27* (1.82)	-0.22 (-0.31)	0.07 (1.15)
R^2	.2973	.259	.2744	.2413	.2668	.2712
N	2,668	2,668	2,668	3,978	3,978	3,978
	AAA (excl. Germany)					
Asset Purchase _t	0.07 (0.64)	0.91 (0.95)	0.02 (0.71)			
R^2	.31	.3021	.2936			
N	977	977	977			

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable:	$\Delta \log \text{Reuse}_t$	$\Delta \text{Reuse Rate}_t$	$\Delta \log(\text{Incoming})_t$	$\Delta \log \text{Reuse}_t$	$\Delta \text{Reuse Rate}_t$	$\Delta \log(\text{Incoming})_t$
Panel B: Dealer-Bond level						
	German			Non-German		
Asset Purchase _t	0.22** (2.42)	1.25** (2.52)	0.14 (1.61)	0.11 (1.40)	0.74* (1.86)	0.09 (1.32)
R ²	.1681	.1698	.1128	.1497	.1656	.1238
N	10,054	10,054	10,054	24,905	24,905	24,905
	AAA			Non AAA		
Asset Purchase _t	0.22*** (2.69)	1.32*** (3.10)	0.13* (1.79)	0.08 (0.78)	0.48 (0.97)	0.09 (1.03)
R ²	.17	.1734	.1164	.1471	.1636	.1252
N	15,169	15,169	15,169	19,689	19,689	19,689
	AAA (excl. Germany)					
Asset Purchase _t	0.12 (0.83)	1.36* (1.82)	0.03 (0.25)			
R ²	.2625	.2714	.2271			
N	5,039	5,039	5,039			

Table 4: Collateral reuse and the effect of asset purchases on repo rates

The table reports the results of a regression of changes in repo rate ($\Delta \text{Repo Rate}_t$) on asset purchases in a bond-time panel at monthly frequency. The regression controls for lagged repo rate, changes in the amount outstanding, dummies for on-the-run and cheapest-to-deliver status, and includes controls for the log change in the securities lending facility (SLF) separately for cash and securities collateral, as well as for the period before and after bond reissuance using dummy variables. In Column (2) we additionally account for the lagged log level of collateral reuse and its interaction with asset purchases, and idem in Column (3) for the lagged reuse rate. Both reuse measures are standardized to have mean zero and unit variance. The full regression model is outlined in Equation (4). The sample consists of German sovereign bonds with a remaining maturity between 1 and 30 years. The sample period is March 2015 - December 2017. t -statistics based on standard errors clustered at the bond level are provided in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)
	Dependent variable: $\Delta \text{Repo Rate}_t$		
Asset Purchase _{t}	-0.93*** (-2.68)	-0.86** (-2.61)	-0.86** (-2.60)
$\Delta \text{Repo Rate}_{t-1}$	-0.44*** (-22.83)	-0.44*** (-22.25)	-0.43*** (-21.42)
Δ Amount outstanding _{t}	0.67** (2.07)	0.67** (2.06)	0.68** (2.10)
Dummy: On-the-run _{t}	-25.12 (-1.65)	-24.76 (-1.66)	-25.69* (-1.67)
Dummy: Cheapest-to-deliver _{t}	0.22 (0.25)	-0.09 (-0.10)	0.07 (0.07)
Δ SLF utilization (sec)	-0.11 (-1.01)	-0.11 (-1.02)	-0.11 (-1.02)
Δ SLF utilization (cash)	-0.22** (-2.20)	-0.22** (-2.22)	-0.22** (-2.23)
Dummy: Before reissuance	0.97 (0.24)	1.34 (0.32)	1.59 (0.36)
Dummy: After reissuance	-7.47 (-0.94)	-7.09 (-0.89)	-8.34 (-1.07)
$\log(\text{Reuse})_{t-1}$		2.17*** (2.80)	
Asset Purchase _{t} (%) $\times$ $\log(\text{Reuse})_{t-1}$		-0.50** (-2.64)	
Reuse rate_{t-1}			1.65*** (2.96)
Asset Purchase _{t} (%) $\times$ Reuse rate_{t-1}			-0.69** (-2.10)
Constant	-1.00 (-1.56)	-1.07* (-1.73)	-1.03 (-1.67)
Fixed effects:			
bond	yes	yes	yes
maturity bucket $\times$ time	yes	yes	yes
R^2	.8552	.8568	.8566
N	1,620	1,620	1,620

Table 5: Collateral reuse and repo market quality

This table presents the results of a regression analysis examining the relationship between market quality measures and collateral reuse, as well as repo specialness, while controlling for factors influencing collateral supply and demand. Panel A displays the results for the average monthly rate of delivery failures, Panel B shows the results for a dummy variable indicating spells of at least five consecutive days of delivery failures for a given security, and Panel C shows the results for repo rate volatility. The *Reuse rate* presented in this table varies between 0 and 1 and has been de-meaned for ease of interpretation. *Dummy: special* is a dummy that equals 1 if a bond has a specialness spread greater than 5 bps and is zero otherwise. In columns (4) - (6) we consider a piece-wise-linear specification, where we estimate separate slopes for high and low reuse rates, where the cutoff point is the 25th percentile (reuse rate = 0.63). We control for the lagged level of fails/volatility, lagged yield, lagged repo rate, bond age, remaining maturity and amount outstanding and on-the-run, cheapest-to-deliver status, reissuance period and the log change in the securities lending facility (SLF) separately for cash and securities collateral. The sample consists of German sovereign bonds with a remaining maturity between 1 and 30 years and the sample period is March 2015 - December 2017. *t*-statistics based on standard errors clustered at the bond level are provided in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Avg. Fails Rate_{<i>t</i>}(%)						
Reuse rate	3.2618*** (3.8403)	3.0984*** (4.0392)	0.1253 (0.2779)			
Dummy: special		0.4772 (1.0133)	0.9014* (1.8390)		0.4161 (0.8940)	0.5142 (0.5361)
Dummy: special × reuse rate			4.2112*** (4.4746)			
Reuse rate low (< P25)				-0.5739 (-0.6091)	-0.6859 (-0.7188)	1.0047 (1.2276)
Reuse rate high (≥ P25)				6.7744*** (4.4770)	6.6041*** (4.6914)	-1.2529 (-0.9177)
Dummy: special × reuse rate low (< P25)						-1.8951 (-1.1206)
Dummy: special × Reuse rate high (≥ P25)						9.2646*** (4.6988)
Controls	yes	yes	yes	yes	yes	yes
Time fixed effects	yes	yes	yes	yes	yes	yes
<i>R</i> ²	.1862	.1868	.1932	.1949	.1953	.2018
N	1,636	1,636	1,636	1,636	1,636	1,636

	(1)	(2)	(3)	(4)	(5)	(6)
Panel B: Spells of at least five consecutive days with failures to deliver						
Reuse rate	0.0951** (2.2079)	0.0924** (2.1054)	-0.0878 (-1.2266)			
Dummy: special		0.0080 (0.2910)	0.0337 (1.1755)		0.0052 (0.1884)	-0.0516 (-0.6739)
Dummy: special $\times$ reuse rate			0.2553*** (3.0491)			
Reuse rate low ($< P25$)				-0.0742 (-0.9815)	-0.0756 (-0.9930)	-0.0540 (-0.4273)
Reuse rate high ($\geq P25$)				0.2501*** (2.9526)	0.2480*** (2.9107)	-0.1409 (-1.0487)
Dummy: special $\times$ reuse rate low ($< P25$)						0.0214 (0.1384)
Dummy: special $\times$ Reuse rate high ($\geq P25$)						0.4490** (2.6070)
Controls	yes	yes	yes	yes	yes	yes
Time fixed effects	yes	yes	yes	yes	yes	yes
R^2	.1211	.1211	.1265	.1249	.125	.1294
N	1,636	1,636	1,636	1,636	1,636	1,636
Panel C: Log Repo Rate Volatility						
Reuse rate	0.5609*** (5.8151)	0.3833*** (5.0466)	0.1239 (1.4575)			
Dummy: special		0.5188*** (7.6163)	0.5558*** (7.8729)		0.5111*** (7.6464)	0.5409*** (5.5885)
Dummy: special $\times$ reuse rate			0.3675*** (2.9761)			
Reuse rate low ($< P25$)				0.0454 (0.4832)	-0.0921 (-0.9870)	0.0539 (0.5186)
Reuse rate high ($\geq P25$)				1.0330*** (6.4791)	0.8237*** (6.0744)	0.2510 (0.9148)
Dummy: special $\times$ reuse rate low ($< P25$)						-0.1805 (-1.1044)
Dummy: special $\times$ Reuse rate high ($\geq P25$)						0.6789** (2.1889)
Controls	yes	yes	yes	yes	yes	yes
Time fixed effects	yes	yes	yes	yes	yes	yes
R^2	.8278	.8399	.8408	.8305	.8422	.8428
N	1,636	1,636	1,636	1,636	1,636	1,636

Table 6: Collateral reuse and mispricing in the secondary bond market

This table reports the results of Regression (6) of yield fitting errors on repo market specialness and Eurosystem asset purchases. Bond controls include the log of amount outstanding, time to maturity squared, and dummy variables for cheapest-to-deliver and on-the-run status. The sample consists of German sovereign bonds with a remaining maturity between 1 and 30 years and the sample period is March 2015 - December 2017 at daily frequency. t -statistics based on standard errors clustered at the bond and day level are provided in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)
Dependent variable:	Δ Yield Fitting Error _{t} (%)			
			Reuse low	Reuse high
Δ Specialness	-0.10*** (-3.36)	-0.08*** (-2.94)	-0.23*** (-3.02)	-0.03 (-0.97)
Asset Purchase		-0.09*** (-2.84)	-0.09* (-1.96)	-0.10** (-2.53)
Δ Specialness $\times$ Asset Purchase		-0.18* (-1.70)	-0.09 (-0.98)	-0.24* (-1.75)
Bond-level controls	yes	yes	yes	yes
Fixed effects:				
time	yes	yes	yes	yes
bond	yes	yes	yes	yes
R^2	.1724	.1728	.1999	.1945
N	37,600	37,600	17,490	20,109

**Internet Appendix accompanying
“Safe asset scarcity, collateral reuse, and market
functioning”**

A Alternative Measures of Reuse

As an alternative to the proportional measure of reuse that we employ in our baseline analysis, one can define an upper bound to the amount of collateral reused as follows (FSB, 2016):

$$Collateral\ reused_{ij}^{upper} = \min(Incoming\ collateral_{ij}, Outgoing\ collateral_{ij}). \quad (7)$$

This measure assumes that a dealer first uses all the incoming collateral of a particular bond before resorting to its outright owned shares.

Alternatively, a lower bound to the amount of collateral reused is given by:

$$Collateral\ reused_{ij}^{lower} = \max((Outgoing\ collateral_{ij} - Own\ assets_{ij}), 0). \quad (8)$$

This measure assumes that a dealer first uses all its outright owned shares of a particular bond before resorting to the incoming collateral.

Consider the following example for illustration of the three reuse measures. Dealer A posts 90 million EUR of a specific bond as collateral. This outgoing collateral can in principle originate from two sources: own assets or incoming collateral. In our example dealer A received 100 million EUR as collateral and owns outright 20 million EUR. Hence, the lower bound of collateral reused is given by $\max((90 - 20), 0) = 70$ million EUR. In this case the dealer first depletes all own holdings (20 million EUR) before using the incoming collateral of which she sources the remaining amount ($90 - 20 = 70$ million EUR). So we know for sure that the dealer reuses 70 million EUR of its incoming collateral. The proportional measure of collateral reuse is given by $(100 / (100 + 20)) \times 90 = 75$ million EUR. Here the dealer obtains collateral proportionally from the two sources, of which the incoming collateral amounts to $100 / (100 + 20) = 83.3\%$. The upper bound of collateral reuse is given by $\min(100, 90) = 90$ million EUR. Here the dealer fully sources her outgoing

collateral with incoming collateral. Relating the amount of collateral reused to the amount of incoming collateral (100 million EUR), the corresponding reuse rates for the lower bound, proportional approach, and upper bound are 70%, 75% and 90%.

Note that the three measures specified in equations (1), (7) and (8) are identical if the dealer has no outright ownership in a particular bond. In this case all the outgoing collateral has to come from incoming collateral. Following the same logic, if the outright ownership becomes small relative to incoming and outgoing collateral the three measures converge.

Table IA.1 provides summary statistics of the three reuse measures. All measures show a similar distribution and a very high correlation between each other. The reuse rate obtained via the proportional approach has a 0.98 correlation with both the reuse rate based on the upper and lower measure. Also the reuse rates based on the upper and lower measure have a correlation of 0.93 between themselves. As a further robustness check we repeat our main analyses from Table 2 using the upper/lower bound approach. The estimation results in Table IA.2 are very similar for all three measures and confirm that our findings do not depend on the assumption of proportional reuse of incoming collateral.

B Instrumental variable approach

In this appendix, we augment our baseline analysis by exploiting exogenous variation implied by the design of the PSPP. Specifically, the PSPP did not allow the purchase of bonds with yields below the deposit facility rate until mid-January 2017.¹ Additionally, there was an initial restriction that limited purchases to bonds with a remaining maturity of more than two years.² These restrictions significantly reduced the number of bonds that were eligible for purchase, particularly for German Bunds. As a result, the Bundesbank

¹The DFR was lowered from negative 20 basis points to negative 30 basis points on December 9, 2015 and from negative 30 to negative 40 basis points on March 16, 2016.

²In January 2017, this threshold was lowered to one year, and the restriction on purchasing assets with yields below the DFR was lifted.

initially could predominantly purchase very long-dated Bund securities. Figure [IA.2](#) plots the minimum maturity of Bund securities eligible for purchase under the PSPP over time.

To exploit this variation in eligibility for the PSPP across bonds, we construct a variable that counts the number of days that a bond was eligible for PSPP purchases. We scale this number with the total number of days since the inception of the PSPP. This relative share thus gives us the fraction of days that a bond is eligible for purchase. The intuition behind this is as follows: purchases of a bond can only occur up to a certain limit on any given day. A bond that is eligible for purchase on only a few days cannot be acquired as frequently as a bond that has been eligible for the entire duration. This effectively directs central banks' purchases toward specific securities.

We use the fraction of eligible days to instrument the PSPP purchases in the first stage of a two-stage least squares regression. The first column of Table [IA.3](#) below shows the results of the first stage. Controlling for various security-specific characteristics, the share of days eligible for the PSPP is a highly significant explanatory variable for the PSPP share, and the F-statistic equal to 205 confirms that it is a strong instrument.

The coefficient of the first stage regression is also economically significant. Comparing a bond that was never eligible to a bond that was eligible throughout results in a $100 \times 0.15 = 15$ percentage points higher PSPP share. In a similar vein, a one standard deviation higher fraction of eligible days corresponds to $38 \times 0.15 = 5.7$ percentage points higher PSPP share.

Having confirmed the validity of our instrument, we then rerun our regression of different proxies of reuse on the instrumented PSPP share. The resulting coefficients are shown in columns 2 to 4 of Table [IA.3](#). They are all highly statistically significant and comparable in magnitude to our baseline OLS results. We take this as evidence that the differential PSPP purchases led to an increased reuse both at the extensive and the intensive margin.

C Sensitivity to Collateral Type

The granularity of our data allows us to study whether the sensitivities to asset purchases further depend on the type of collateral. We conduct two such analyses, which extend on the regressions in Table 2.

First, in Table IA.4 we consider the (additional) sensitivity of on-the-run bonds by including an interaction term of asset purchases with the on-the-run dummy. The estimation coefficient for this interaction term is only significant at the 10% level in Specification (4) where we regress the amount of collateral reused for the extended sample of European bonds. This suggests that European on-the-run bonds are more sensitive to purchases than off-the-run bonds.

Second, we distinguish between collateral with different remaining maturities in Table IA.5. That is, we repeat the sensitivity analysis of the reuse rate for subsets of shorter-dated collateral (bonds with a remaining maturity of between 1 and 5 years, medium-dated collateral (remaining maturity between 5 and 10 years) and longer-dated collateral (remaining maturity between 10 and 30 years). For both European and German collateral a “term-structure” of sensitivities to asset purchases across these maturity bands is discernible, with shorter- and medium-dated collateral being more sensitive to asset purchases than longer-dated collateral.

D Economic magnitude of the collateral reuse channel

We have shown that dealers react to scarcity-inducing purchases primarily via the intensive margin, that is by adjusting their reuse rate. To highlight the economic significance of these adjustments we perform the following exercise. Given a reduction in the supply of collateral, how much additional collateral do dealers need to provide through collateral reuse in order to maintain a constant amount of collateral in the market?

For a given base amount of collateral that is available, we can compute the effective amount of available collateral it is able to support as follows³:

$$\begin{aligned} \text{effective amount} &= \text{base amount} \times \sum_{n=0}^{\infty} \text{reuse rate}^n \\ &= \frac{\text{base amount}}{1 - \text{reuse rate}}. \end{aligned} \tag{9}$$

The intuition behind Equation (9) is the following. Suppose bank A uses a certain amount of a bond as collateral in a transaction with bank B. This collateral is sourced from its outright holdings and we refer to it as the base amount of collateral. Bank B has access to this amount and reuses part of this collateral in another transaction with bank C. At this point the effective amount of collateral available is equal to the sum of the base amount, the one received by bank B and the amount received by bank C. As this series of reuses goes to infinity, it can be approximated as a geometric sum, yielding the second identity of Equation (9), see also [Bottazzi et al. \(2012\)](#).

Given our estimated reuse rates, what would this imply for the total collateral available? To answer this question, we calibrate Equation (9) to our data. We assume a reuse rate of 80.7%, which is the median value in the dealer-security panel used for estimating Table 2. Hence, at the given value of reuse rate one unit of a bond supports 5.18 times as much collateral in the market.

Given a reduction in the supply of collateral, how do dealers need to adjust their reuse rate in order to maintain constant the effective amount of collateral? Equation (9) implies that for a reduction in the base amount by 1%, the new reuse rate in our example needs to be 80.9%, i.e. an increase by 0.2 percentage points. This is considerably lower than the estimated coefficient in Table 2, Column (1). An asset purchase of 1% of the amount outstanding increases the reuse rate by 1.25 percentage points. This conclusion is robust at different levels of reuse and taking into account haircuts, see Table IA.6 in the Internet

³We thank Toomas Laarits for this suggestion.

Appendix. Hence, at first sight dealers seem to overcompensate the collateral reduction through collateral reuse.

One potential explanation for this discrepancy is that a purchase of one percent of the amount outstanding actually corresponds to a substantially larger depletion of the pool of collateral that can be accessed by dealers. This could be the case if central banks buy disproportionately from holders that would otherwise supply these assets as collateral. Hence a purchase of one percent of the amount outstanding may actually correspond to a reduction in the effective amount of collateral available to reuse that is considerably larger. Consistent with this notion, [Koijsen et al. \(2021\)](#) find that euro area banks, which generally supply their holdings as collateral, are the second largest net seller of euro area government bonds (after foreign investors) to the Eurosystem. They reduced their holdings by 470 billion EUR from the first quarter of 2015 to the last quarter of 2017, corresponding to 25% of purchases. In contrast, insurance companies and pension funds, which are generally less likely to supply collateral to the market ([Duffie, 1996](#)), increased their holdings over the same period.⁴

E Summary Statistics for High and Low Reuse Bund Securities

In Section 6, we distinguish between high and low reuse bonds as those above and below the 25th percentile of the reuse rate distribution across securities and time periods. To better understand the security characteristics which drive this distinction, Table [IA.8](#) below reports summary statistics for the two subsamples. They show that reuse is very similar across bonds with different remaining time to maturity and for “on-the-run securities”. That said, there are sizable and significant differences with respect to the cheapest-to-deliver status and the specialness spread (bottom two rows in the table). In contrast, the

⁴Another explanation could be that reuse chains are not infinitely long. That said, if we assume that each unit of collateral is reused only five (ten) times, i.e. if we truncate the sum in Equation (9) at $n = 5$ ($n = 10$), we still obtain a multiplier of 3.70 (4.69) in the example above. When we assume that collateral is reused only about twice the calculated estimate is comparable to the empirically observed value. However we deem such short chains unlikely.

reuse rate is significantly higher for bonds which are cheapest-to-deliver in the futures market and which have higher specialness spreads. There is also a statistically significant but economically small difference with respect to the amount outstanding. We further note that since reissuances are very common for Bunds, on-the-run effects are much less important in the Bund market compared to U.S. Treasuries ([Ejsing and Sihvonen, 2009](#)). Accordingly, the on-the-run status does not appear to be very relevant for banks' decision to reuse a security.

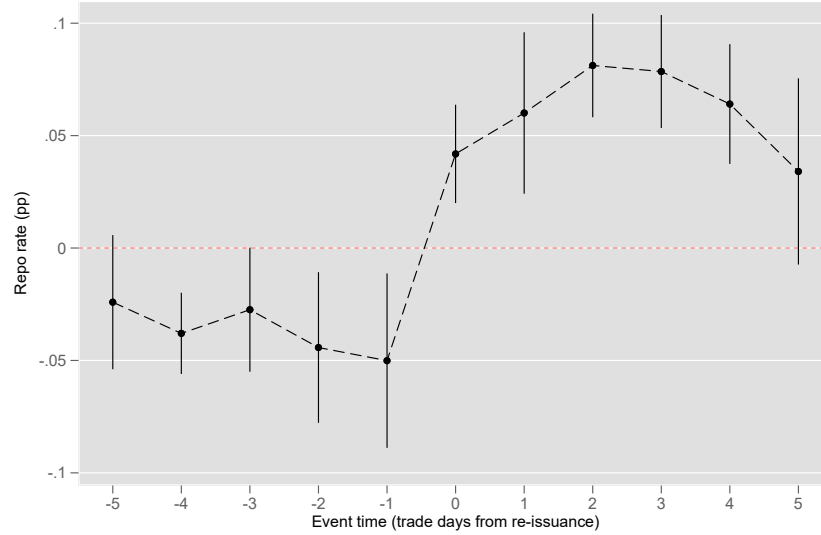


Figure IA.1: Repo rates around reissuance events

This figure illustrates the repo rate dynamics of German sovereign bonds in the vicinity of reissuance events. The analysis is based on German sovereign bonds with remaining maturities ranging from 1 to 30 years in the time period March 2015 until December 2017. To capture the impact of reissuance, we regress the repo rate on daily dummies spanning the period around the reissuance date (from $t = -5$ to $t = +5$), while controlling for security fixed effects. The reference period for comparison is defined as the days from $t = -8$ to $t = -6$. Standard errors are clustered at the security level, and the whiskers represent 90% confidence intervals.

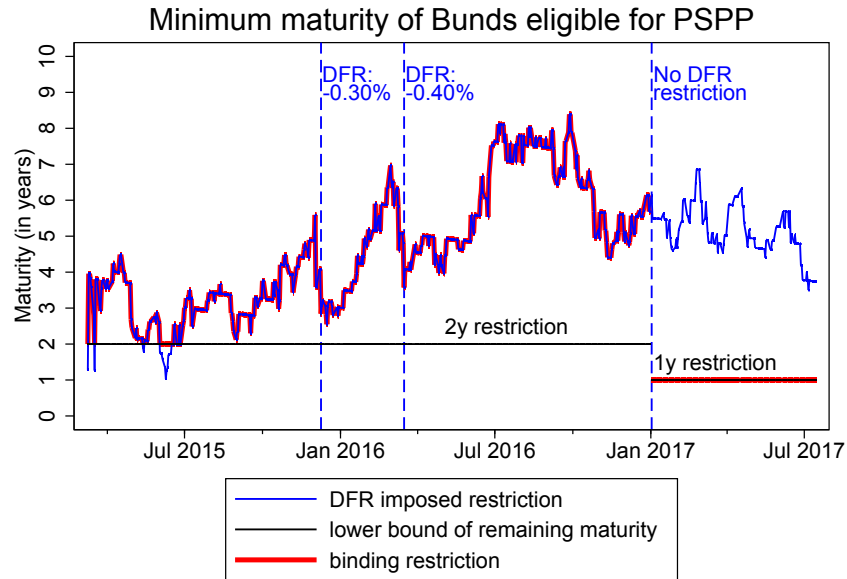


Figure IA.2: Minimum maturity of Bunds eligible for PSPP

This figure illustrates the minimum maturity of German sovereign bonds eligible for the PSPP. Prior to January 2017, the purchase of assets with yields below the Deposit Facility Rate (DFR) was prohibited. The blue line represents the minimum remaining maturity for which Bunds could be purchased. Initially, a maturity restriction limited purchases to bonds with a remaining maturity of more than 2 years. In January 2017, this threshold was reduced to 1 year. Simultaneously, the restriction on purchasing assets with yields below the DFR was lifted. The sample period spans from March 2015 to July 2017.

Table IA.1:
Descriptive statistics: reuse rates

This table shows summary statistics and correlations of collateral reuse rates, employing the three different measures for collateral reuse activity. The sample consists of the security-level panel of European sovereign bonds with remaining maturity between 1 and 30 years. The sample period is 2008 - 2017 at quarterly frequency.

Row	Variable	Mean	Std. dev.	Percentiles			Correlation		
				25th	50th	75th	(1)	(2)	(3)
(1)	$reuse\ rate^{lower}\ (\%)$	70.4	29.0	59.8	80.0	92.8	1		
(2)	$reuse\ rate^{prop.}\ (\%)$	72.3	28.2	63.3	81.4	93.3	0.98	1	
(3)	$reuse\ rate^{upper}\ (\%)$	74.6	28.6	66.1	84.6	94.9	0.93	0.98	1

Table IA.2:**Asset purchases and collateral reuse: intensive and extensive margin****Robustness check: Using alternative reuse measures.**

This table provides a robustness check to the analysis of Table 2 using the upper- and lower-bound reuse as dependent variable in the regression instead. The dependent variable is changes in logarithmic amount of collateral reused ($\Delta \log(\text{Reuse})_t$) in specifications (1) - (3), and changes in reuse rate ($\Delta \text{Reuse Rate}_t$) in specifications (4) - (6). In specifications (1) and (4) we employ the lower bound measure for reuse instead, and in specifications (3) and (6) the upper bound measure. Panel A is at the bond level and specifications (2) and (5) are identical to specifications (1) and (2) in Table 2, respectively. Panel B is at the dealer-bond level and specifications (2) and (5) therein are identical to specifications (4) and (5) in Table 2, respectively. The sample consists of German sovereign bonds with a remaining maturity between 1 and 30 years. The sample period is March 2015 - December 2017. t -statistics based on clustered standard errors are provided in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: bond level						
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable:	$\Delta \log(\text{Reuse})_t$			$\Delta \text{Reuse Rate}_t$		
Reuse computation:	Lower bound	Prop. measure	Upper bound	Lower bound	Prop. measure	Upper bound
Asset Purchase _{<i>t</i>} (%)	0.13** (2.30)	0.13** (2.21)	0.13** (2.19)	1.51*** (2.75)	1.63*** (3.10)	1.73*** (3.22)
Δ Amount outstanding _{<i>t</i>}	0.01 (0.80)	0.01 (1.12)	0.01 (1.17)	-0.20* (-1.85)	-0.16 (-1.56)	-0.12 (-1.22)
Dummy: On the run _{<i>t</i>}	0.29 (1.07)	0.15 (0.55)	0.11 (0.41)	9.99** (2.21)	7.52* (1.77)	4.89 (1.22)
Dummy: Cheapest-to-deliver _{<i>t</i>}	-0.06 (-0.72)	-0.05 (-0.63)	-0.05 (-0.62)	-1.40 (-0.83)	-0.95 (-0.59)	-0.81 (-0.52)
Constant	-0.10 (-1.49)	-0.10 (-1.45)	-0.10 (-1.44)	-1.16** (-2.08)	-1.29** (-2.39)	-1.37** (-2.55)
Fixed effects:						
bond	yes	yes	yes	yes	yes	yes
R^2	.2394	.2445	.2428	.2114	.2114	.205
N	1,691	1,691	1,691	1,691	1,691	1,691

Panel B: dealer-bond level

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable:	$\Delta \log(\text{Reuse})_t$			$\Delta \text{Reuse Rate}_t$		
Reuse computation:	Lower bound	Prop. measure	Upper bound	Lower bound	Prop. measure	Upper bound
Asset Purchase _t (%)	0.23** (2.48)	0.22** (2.42)	0.22** (2.42)	1.11** (2.23)	1.25** (2.52)	1.32*** (2.63)
Δ Amount outstanding _t	0.02 (1.15)	0.03* (1.65)	0.03* (1.74)	0.03 (0.36)	0.05 (0.60)	0.08 (1.05)
Dummy: On the run _t	0.62 (0.74)	0.45 (0.59)	0.40 (0.52)	5.99 (1.41)	5.21 (1.27)	3.52 (0.88)
Dummy: Cheapest-to-deliver _t	-0.06 (-0.19)	-0.12 (-0.33)	-0.12 (-0.34)	-0.78 (-0.38)	-0.86 (-0.42)	-0.75 (-0.36)
Constant	-0.20 (-1.63)	-0.20* (-1.65)	-0.20* (-1.65)	-0.93 (-1.35)	-1.10 (-1.59)	-1.18* (-1.68)
Fixed effects:						
dealer $\times$ time	yes	yes	yes	yes	yes	yes
dealer $\times$ bond	yes	yes	yes	yes	yes	yes
maturity bucket $\times$ country $\times$ time	yes	yes	yes	yes	yes	yes
R^2	.1586	.1681	.1678	.1639	.1698	.1717
N	10,054	10,054	10,054	10,054	10,054	10,054

Table IA.3: Asset purchases and collateral reuse: IV-approach

The table presents the effects of asset purchases (measured as the level share) on reuse, the reuse rate, and the sourcing of incoming collateral. In the first-stage regression, we instrument the level share of PSPP ownership using the fraction of days a bond was eligible for asset purchases. This eligibility fraction is determined by a rule-based criterion: whether a bond's yield was below (ineligible) or above (eligible) the deposit facility rate. This rule remained in effect until January 2017. Column (1) shows the first-stage regression, columns (2) - (4) the second-stage regressions. The sample period is March 2015 - December 2016. *t*-statistics based on standard errors clustered by security are provided in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Dependent variable:	1st stage PSPP share	2nd stage reuse	2nd stage reuse rate	2nd stage incoming collateral
Share of days eligible (%)	0.15*** (14.33)			
PSPP share		0.07*** (3.32)	1.34*** (4.15)	0.04** (2.16)
Maturity remaining	-0.33*** (-7.54)	-0.03** (-2.13)	0.17 (0.81)	-0.03*** (-2.90)
Age	0.10* (1.77)	-0.04* (-1.80)	-1.01*** (-3.83)	-0.02 (-1.08)
log amt. outstanding	0.88 (0.84)	0.50 (1.39)	2.71 (0.48)	0.44 (1.61)
Dummy: on-the-run	0.72 (0.32)	-0.82 (-1.41)	1.88 (0.22)	-0.62 (-1.23)
Elig. Futures delivery	-1.22 (-1.37)	0.19 (1.15)	6.91** (2.45)	0.05 (0.33)
Dummy: Cheapest to deliver	-0.97 (-0.86)	0.12 (0.51)	4.93 (1.47)	0.10 (0.57)
F-statistic	205.4			
R^2		.1356	.1345	.1529
N	1,019	1,019	1,019	1,019

Table IA.4:**Asset purchases and collateral reuse: intensive and extensive margin****Robustness check: Sensitivity of on-the-run bonds.**

This table provides an extension to the analysis of Table 2 by estimating the sensitivity of on-the-run bonds to asset purchases. The table reports the results of a regression of changes in collateral reuse on asset purchases in a dealer-bond-time panel at monthly frequency. In addition to the specification of Table 2 we include an interaction term ($\text{Asset Purchase}_t (\%) \times \text{Dummy: On the run}_t$) that captures the additional sensitivity of on-the-run bonds to asset purchases. In specifications (1) and (4) the dependent variable is changes in logarithmic amount of collateral reused ($\Delta \log \text{Reuse}_t$), and in specifications (2) and (5) changes in reuse rate ($\Delta \text{Reuse Rate}_t$). The dependent variable in specifications (3) and (6) is changes in the logarithmic amount of incoming collateral ($\Delta \log(\text{Incoming})_t$). We account also for changes in the amount outstanding and control for on-the-run and cheapest-to-deliver status. The sample consists of German sovereign bonds in specifications (1) - (3), and of European sovereign bonds in specifications (4) - (6). The remaining maturity is between 1 and 30 years. The sample period is March 2015 - December 2017. t -statistics based on clustered standard errors are provided in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	German collateral			European collateral		
Dependent variable:	$\Delta \log(\text{Reuse})_t$	$\Delta \text{Reuse Rate}_t$	$\Delta \log(\text{Incoming})_t$	$\Delta \log(\text{Reuse})_t$	$\Delta \text{Reuse Rate}_t$	$\Delta \log(\text{Incoming})_t$
Asset Purchase _{<i>t</i>} (%)	0.03 (1.23)	1.21** (2.10)	-0.00 (-0.07)	0.09 (1.64)	0.53 (1.31)	0.03 (1.25)
Asset Purchase _{<i>t</i>} (%) $\times$ Dummy: On the run _{<i>t</i>}	0.28 (1.63)	1.17 (1.09)	0.28 (1.61)	0.28* (1.96)	1.08 (1.35)	0.16 (1.35)
Δ Amount outstanding _{<i>t</i>}	0.01 (0.87)	-0.17 (-1.60)	0.01 (1.61)	-0.01 (-1.03)	-0.21*** (-3.14)	0.01* (1.75)
Dummy: On the run _{<i>t</i>}	-0.46 (-0.82)	4.96 (1.04)	-0.65 (-1.18)	-0.11 (-0.33)	4.08* (1.72)	-0.26 (-0.93)
Dummy: Cheapest-to-deliver _{<i>t</i>}	 (-0.15)	-0.01 (-0.49)	-0.78 (0.23)	0.02 (-0.39)	-0.07 (0.13)	0.21 (-1.60)
Constant	-0.02 (-0.66)	-0.97* (-1.70)	-0.00 (-0.00)	0.02 (0.44)	-0.14 (-0.46)	-0.01 (-0.57)
Fixed effects:						
bond	yes	yes	yes	yes	yes	yes
maturity bucket $\times$ country $\times$ time	yes	yes	yes	yes	yes	yes
R^2	.2638	.2122	.2979	.2466	.2626	.2695
N	1,691	1,691	1,691	6,646	6,646	6,646

Table IA.5:
Asset purchases and collateral reuse: intensive and extensive margin
Robustness check: Comparison across bond maturities.

This table provides an extension to the analysis of Table 2 by separating shorter- and longer-dated bonds in their sensitivity to asset purchases. The table reports the results of a regression of changes in collateral reuse on asset purchases in a dealer-bond-time panel at monthly frequency. In Panel A the dependent variable is changes in logarithmic amount of collateral reused ($\Delta \log \text{Reuse}_t$), and in Panel B changes in reuse rate ($\Delta \text{Reuse Rate}_t$). The dependent variable in Panel C is changes in the logarithmic amount of incoming collateral ($\Delta \log(\text{Incoming})_t$). Specification (1) in each of the three panels repeats specifications (1) - (3) of Table 2, respectively, and is based on all bonds with a remaining maturity between 1 and 30 years. In specifications (2) and (6) the remaining maturity is between 1 and 5 years, while in specifications (3) and (7) it is between 5 and 10 years. The sample in specifications (4) and (8) is for bonds with a remaining maturity between 10 and 30 years. The sample consists of German sovereign bonds in specifications (1) - (4), and in specifications (5) - (8) of European sovereign bonds as described in Section 4.2. We account also for changes in the amount outstanding and control for on-the-run and cheapest-to-deliver status. The sample period is March 2015 - December 2017. t -statistics based on clustered standard errors are provided in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: changes in logarithmic amount of collateral reused								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	German collateral				European collateral			
Remaining maturity:	1 – 5 years		5 – 10 years	10+ years	1 – 5 years		5 – 10 years	10+ years
Dependent variable:	$\Delta \log \text{Reuse}_t$							
Asset Purchase _{<i>t</i>} (%)	0.13** (2.21)	0.23** (2.30)	0.15** (2.56)	0.06 (1.08)	0.17*** (2.80)	0.22* (1.96)	0.10 (1.50)	0.26 (1.59)
Δ Amount outstanding _{<i>t</i>}	0.01 (1.12)	0.03* (1.81)	-0.01 (-0.45)	0.07 (1.23)	-0.01 (-0.70)	0.00 (0.10)	0.00 (0.20)	-0.08** (-2.02)
Dummy: On the run _{<i>t</i>}	0.15 (0.55)	-0.23 (-0.45)	0.51 (1.10)	0.00 (.)	0.40 (1.47)	0.07 (0.10)	-0.11 (-0.28)	0.15 (0.47)
Dummy: Cheapest-to-deliver _{<i>t</i>}	-0.05 (-0.63)	-0.32* (-1.72)	0.05 (0.53)	-0.07 (-0.86)	-0.09 (-0.54)	-0.24 (-0.65)	-0.39 (-1.39)	0.03 (0.41)
Constant	-0.10 (-1.45)	-0.11 (-1.20)	-0.17** (-2.67)	-0.04 (-0.87)	-0.04 (-0.78)	-0.06 (-0.99)	0.03 (0.45)	0.03 (0.23)
Fixed effects:								
bond	yes	yes	yes	yes	yes	yes	yes	yes
maturity bucket $\times$ country $\times$ time	yes	yes	yes	yes	yes	yes	yes	yes
<i>R</i> ²	.2445	.3226	.1404	.2712	.246	.3007	.1963	.213
N	1,691	791	514	381	6,646	2,807	2,285	1,538

Panel B: changes in reuse rate

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	German collateral				European collateral			
Remaining maturity:	1 – 5 years		5 – 10 years	10+ years	1 – 5 years		5 – 10 years	10+ years
Dependent variable:	$\Delta \text{Reuse Rate}_t$							
Asset Purchase _t (%)	1.63*** (3.10)	1.64** (2.51)	3.54*** (3.60)	0.41 (0.15)	0.86** (2.27)	1.11* (1.83)	1.45** (2.54)	-0.71 (-0.65)
Δ Amount outstanding _t	-0.16 (-1.56)	0.01 (0.10)	-0.68** (-2.65)	0.99 (0.70)	-0.19*** (-2.98)	-0.07 (-0.70)	-0.29*** (-3.05)	-0.55** (-2.45)
Dummy: On the run _t	7.52* (1.77)	2.46 (0.33)	21.34*** (2.79)	0.00 (.)	6.08*** (3.45)	4.57 (1.28)	5.59** (2.02)	0.61 (0.24)
Dummy: Cheapest-to-deliver _t	-0.95 (-0.59)	-4.74 (-1.26)	2.62 (1.12)	-1.16 (-0.40)	0.11 (0.07)	-1.44 (-0.41)	-0.70 (-0.29)	-1.66 (-0.65)
Constant	-1.29** (-2.39)	-0.53 (-0.88)	-4.17*** (-3.48)	-0.43 (-0.17)	-0.36 (-1.21)	-0.37 (-1.07)	-0.72 (-1.27)	1.84* (1.78)
Fixed effects:								
bond	yes	yes	yes	yes	yes	yes	yes	yes
maturity bucket×country×time	yes	yes	yes	yes	yes	yes	yes	yes
R ²	.2114	.1888	.2619	.2839	.2624	.2876	.2544	.2532
N	1,691	791	514	381	6,646	2,807	2,285	1,538

Panel C: changes in the logarithmic amount of incoming collateral

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	German collateral				European collateral			
Remaining maturity:	1 – 5 years		5 – 10 years	10+ years	1 – 5 years		5 – 10 years	10+ years
Dependent variable:	$\Delta \log(\text{Incoming})_t$							
Asset Purchase _t (%)	0.10*	0.21**	0.08	0.05	0.08**	0.15**	0.09**	0.05
	(1.72)	(2.07)	(1.58)	(1.10)	(2.24)	(2.17)	(2.09)	(0.50)
Δ Amount outstanding _t	0.02*	0.03*	0.01	0.05	0.02*	0.03*	0.01	0.00
	(1.82)	(1.99)	(0.49)	(1.79)	(1.91)	(1.66)	(1.11)	(0.10)
Dummy: On the run _t	-0.04	-0.36	0.05	0.00	0.03	-0.31	-0.15	0.09
	(-0.17)	(-0.74)	(0.12)	(.)	(0.14)	(-1.17)	(-0.46)	(0.37)
Dummy: Cheapest-to-deliver _t	-0.02	-0.22	-0.01	-0.05	-0.28*	-0.12	-0.62	-0.05
	(-0.28)	(-1.34)	(-0.09)	(-0.64)	(-1.69)	(-0.69)	(-1.65)	(-0.54)
Constant	-0.08	-0.11	-0.09*	-0.04	-0.04	-0.07	-0.05	-0.02
	(-1.14)	(-1.18)	(-1.75)	(-0.85)	(-1.38)	(-1.63)	(-1.03)	(-0.22)
Fixed effects:								
bond	yes	yes	yes	yes	yes	yes	yes	yes
maturity bucket $\times$ country $\times$ time	yes	yes	yes	yes	yes	yes	yes	yes
R^2	.2719	.3882	.1474	.159	.2689	.3877	.1973	.2624
N	1,691	791	514	381	6,646	2,807	2,285	1,538

Table IA.6:**Adjustment to reuse rates necessary to compensate supply reductions.**

This table provides a robustness check to the analysis in Section D using a wide set of potential parameters. Taking into account a haircut HC that is applied each time that collateral is reused, Equation (9) becomes $effective\ amount = base\ amount \times (1 - HC) \times \sum_{n=0}^{\infty} (reuse\ rate \times (1 - HC))^n = base\ amount \times \frac{(1-HC)}{1-reuse\ rate \times (1-HC)}$. Specifically we compute the reuse rate $reuse\ rate'$ that is necessary to compensate for a reduction by one percent in collateral supply ($base\ amount$) given the initial $reuse\ rate$ and haircut. $\Delta reuse\ rate$ gives the corresponding increase in the reuse rate in percentage points. The mean (median) haircut for our sample as reported in the ECB's eligible assets database is 2% (3.3%).

$reuse\ rate\ (\%)$	haircut $(\%)$	$reuse\ rate'\ (\%)$	$\Delta reuse\ rate\ (\%)$
20.00	0.00	20.80	0.80
40.00	0.00	40.60	0.60
60.00	0.00	60.40	0.40
70.00	0.00	70.30	0.30
80.00	0.00	80.20	0.20
80.70	0.00	80.89	0.19
90.00	0.00	90.10	0.10
95.00	0.00	95.05	0.05
99.00	0.00	99.01	0.01
20.00	3.00	20.83	0.83
40.00	3.00	40.63	0.63
60.00	3.00	60.43	0.43
70.00	3.00	70.33	0.33
80.00	3.00	80.23	0.23
80.70	3.00	80.92	0.22
90.00	3.00	90.13	0.13
95.00	3.00	95.08	0.08
99.00	3.00	99.04	0.04
20.00	5.00	20.85	0.85
40.00	5.00	40.65	0.65
60.00	5.00	60.45	0.45
70.00	5.00	70.35	0.35
80.00	5.00	80.25	0.25
80.70	5.00	80.95	0.25
90.00	5.00	90.15	0.15
95.00	5.00	95.10	0.10
99.00	5.00	99.06	0.06

Table IA.7:**The Effect of asset purchases on Repo rates****Robustness check: Extended sample of bonds.**

This table provides a robustness check to the analysis of Table 4 using a more general universe of bonds. We consider domestic and non-domestic sovereign bonds for which we observe a reuse activity comparably to domestic collateral. Specifically, we standardize the aggregate amount of collateral reuse of all dealers in our sample by dividing it through the total amount outstanding. For a bond to be included in the sample, we require it to be greater or equal to the 20th percentile of the domestic collateral distribution. The table reports the results of a regression of changes in repo rate ($\Delta \text{Repo Rate}_t$) on asset purchases in a bond-time panel at monthly frequency. We account also for changes in the amount outstanding and control for on-the-run and cheapest-to-deliver status. In specification (2) we additionally account for the lagged level of collateral reuse normalized by outright ownership in the same bond, and its interaction with asset purchases, and idem in specification (3) for the lagged reuse rate. Both reuse measures are standardized to have unit variance. The full regression model is outlined in Equation (4). The remaining maturity of all bonds is between 1 and 30 years. The sample period is March 2015 - December 2017. t -statistics based on standard errors clustered at the bond level are provided in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)
Dependent variable: $\Delta \text{Repo Rate}_t$			
Asset Purchase _t (%)	-0.94*** (-2.83)	-1.01*** (-3.00)	-0.88*** (-2.95)
$\Delta \text{Repo Rate}_{t-1}$	-0.44*** (-20.91)	-0.45*** (-22.13)	-0.44*** (-19.25)
Δ Amount outstanding _t	0.40* (1.79)	0.39 (1.62)	0.40* (1.75)
Dummy: On-the-run _t	-15.14 (-1.46)	-15.38 (-1.44)	-15.25 (-1.43)
Dummy: Cheapest-to-deliver _t	-0.10 (-0.12)	-0.19 (-0.24)	-0.26 (-0.29)
$\log(\text{Reuse}/\text{Outright ownership})_{i,t-1}$		1.35*** (2.85)	
Asset Purchase _t (%) $\times$ $\log(\text{Reuse}/\text{Outright ownership})_{i,t-1}$		-0.61** (-2.31)	
Reuse rate _{t-1}			1.23** (2.43)
Asset Purchase _t (%) $\times$ Reuse rate _{t-1}			-0.47 (-1.25)
Constant	-1.56*** (-3.26)	-1.54*** (-3.19)	-1.57*** (-3.47)
Fixed effects:			
bond	yes	yes	yes
maturity bucket $\times$ time	yes	yes	yes
R^2	.8437	.8458	.8444
N	1,775	1,731	1,775

Table IA.8: Comparison of high and low-reuse securities

The table provides a comparison of summary statistics across securities with high vs. low reuse rate which have been used in Table 5. The cutoff point is the 25th percentile of the reuse rate distribution which corresponds to a reuse rate of 0.63. The analysis covers the sample period from March 2015 to December 2017.

	Reuse low	Reuse high	Diff	t-value
Maturity remaining (years)	7.82	7.70	0.12	0.29
Amount outstanding (EUR bn)	16.30	18.15	-1.85	-8.34
Dummy: On-the-run	0.01	0.01	-0.00	-0.51
Dummy: CTD	0.04	0.11	-0.08	-4.65
Specialness spread (ppt)	0.15	0.25	-0.10	-9.50
